



asking those who feel it

**local and Indigenous knowledge (ILK)
on climate change**

Day 2 (6 December)

Jakob Steiner

Course content

5 December (1400 – 1800)

Pre course evaluation - general introduction and history – guest lecture (Arctic)- introduction to writing assignment

6 December (0900 – 1700)

Guest lecture (Himalaya) – Climate Science– interactive team work - International efforts to integrate ILK

7 December (0900 – 1700)

ILK governance and ethics – integrating ILK with conventional data - interactive team work - group presentation and discussion

6 December / Day 2

Welcome to our second session 😊

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ILK in the Arctic

- longer standing of appreciation and discussion of ILK in research (from the turn of the Millenium)
- in parallel greater representation of Indigenous knowledge holders in intergovernmental agencies
- less separate as many scientists are also Indigenous knowledge holders
- focus on environmental change, ecology, health and archeology
- in North America research in Indigenous lar already legally subject to approval



ALEUT INTERNATIONAL ASSOCIATION

www.aleut-international.org



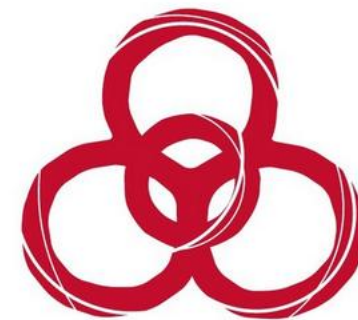
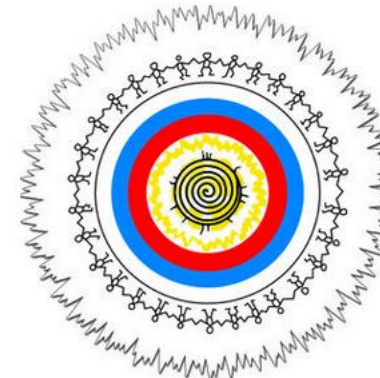
ARCTIC ATHABASKAN COUNCIL

www.arcticathabaskancouncil.com



GWICH'IN COUNCIL INTERNATIONAL

www.gwichincouncil.com

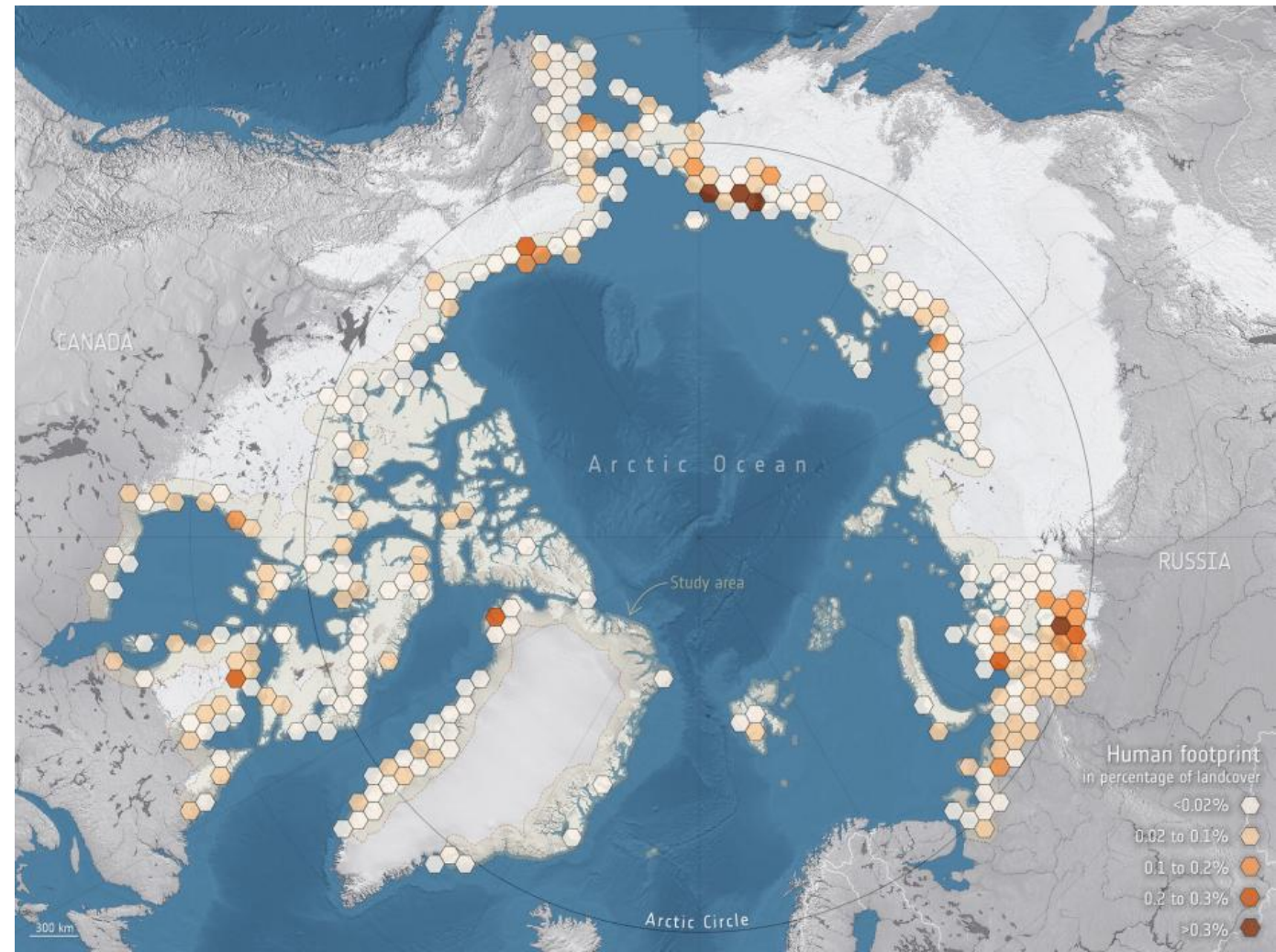


e.g. Sjöberg et al. (2015)

www.arcticpeoples.com

ILK in the Arctic

- governance (especially related to food production and infrastructure) critical
- ca 4 million people (344 million in mountains is the lowest estimate)



e.g. Bartsch et al. (2021)

Variables collected (largely factual knowledge)

- collecting such data needs

- a) Time
- b) Language skills
- c) Capacity to listen

Riedlinger and Berkes (2001)

Table 2. Examples of recent local environmental change related to climate, as described by Inuvialuit in Sachs Harbour, western Arctic.

Phenomena	Observed change
Sea ice	Less/no multiyear ice in July and August; more open water (and 'rougher' water) More ice movement than before Not able to see the permanent pack ice to the west Ice breaks up earlier and freezes up later Rate of ice break-up has increased Annual ice in harbour is weaker, thinner (not safe) Less and thinner landfast ice (shore ice) Changes in distribution and extent of local pressure ridges Leads (openings in ice) farther away from shore Ice pans no longer push up on shore Open water in winter is closer than before Changes in ice colour and texture
Permafrost	Land subsiding in some places Increasing slumping and landslides, both inland and along the coast Exposed ground ice (ice lenses) on hillsides Increased depth of active layer in spring Increased rate of melt in spring Water and 'ice pebbles' in the ground rather than ice in some places Increased puddles, ponds, water in pools on the flat land Pingos decreasing in size More mud on the land
Changing seasons	Longer, warmer summers More rain and wind in the summer Melts faster in the spring than it used to Spring comes earlier now than it used to Shorter, warmer winters August is warm month now; used to be 'cooling off month' Autumn comes later now than before Seasonal change is more unpredictable now
Fish and wildlife	Two species of Pacific salmon caught near the community Increased numbers of <i>Coregonus sardinella</i> (least cisco) Fewer polar bears in area because of less ice Increasing occurrence of 'skinny' seal pups at spring break-up Observations of robins; previously unknown small birds Increased forage availability for caribou and muskox Changes to timing of intra-island caribou migration
Weather	More rain in the fall; rain that should have been snow Different kind of snow, fewer rough drifts Longer duration of 'hot' days, now a whole week rather than 1–2 days Thunder and lightning – none seen before Weather more unpredictable Changes in wind velocity and direction; more intense wind storms

Table 1. Assessment of Respectful and Ethical Inclusion of TEK in the UNFCCC, the Kyoto Protocol, the AR4, and the ACIA

Indicator	UNFCCC	Kyoto Protocol	AR4	ACIA
Terminology consistent with indigenous peoples practices	No	No	Select chapters	Select chapters
Knowledge presented indigenous peoples' knowledge	No	No	Select chapters	Select chapters
Knowledges treated as situated and unique	No	No	Select chapters	Select chapters
Value assigned to indigenous knowledges	No	No	Select chapters	Select chapters
Inclusion of indigenous peoples' in research process	No	No	No	Select chapters
Indigenous peoples' generated knowledge	No	No	No	Select chapters

United Nations Framework Convention on Climate Change (UNFCCC)

IPCC (AR4)

Arctic Climate Impact Assessment (ACIA) - <https://acia.amap.no/>

Smith and Sharp (2012)

Fig. 2. Responses on how ECRs view their own and Indigenous peoples' understanding of different aspects relevant for Arctic research (a). Selected quotes demonstrating the expectations of ECRs when involving Indigenous people in research (b).

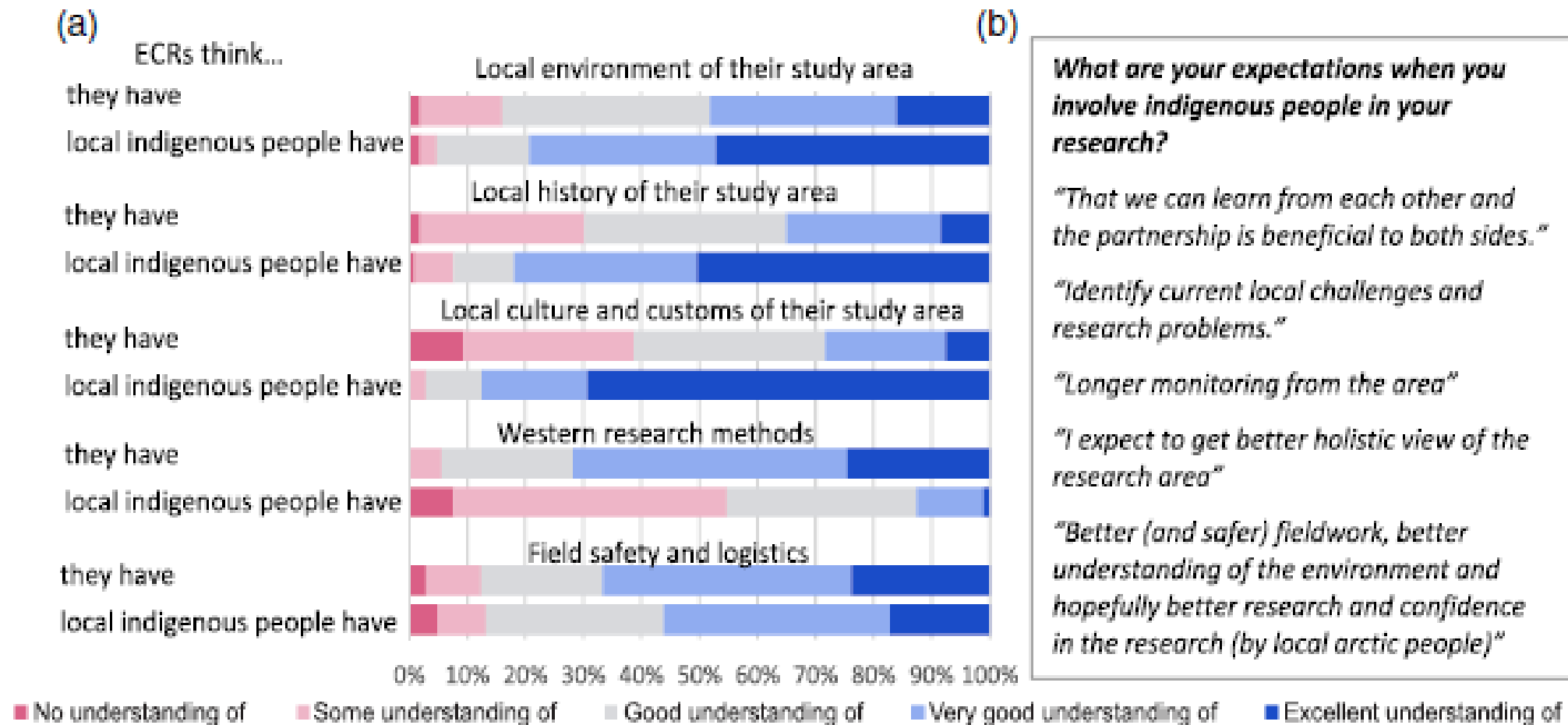
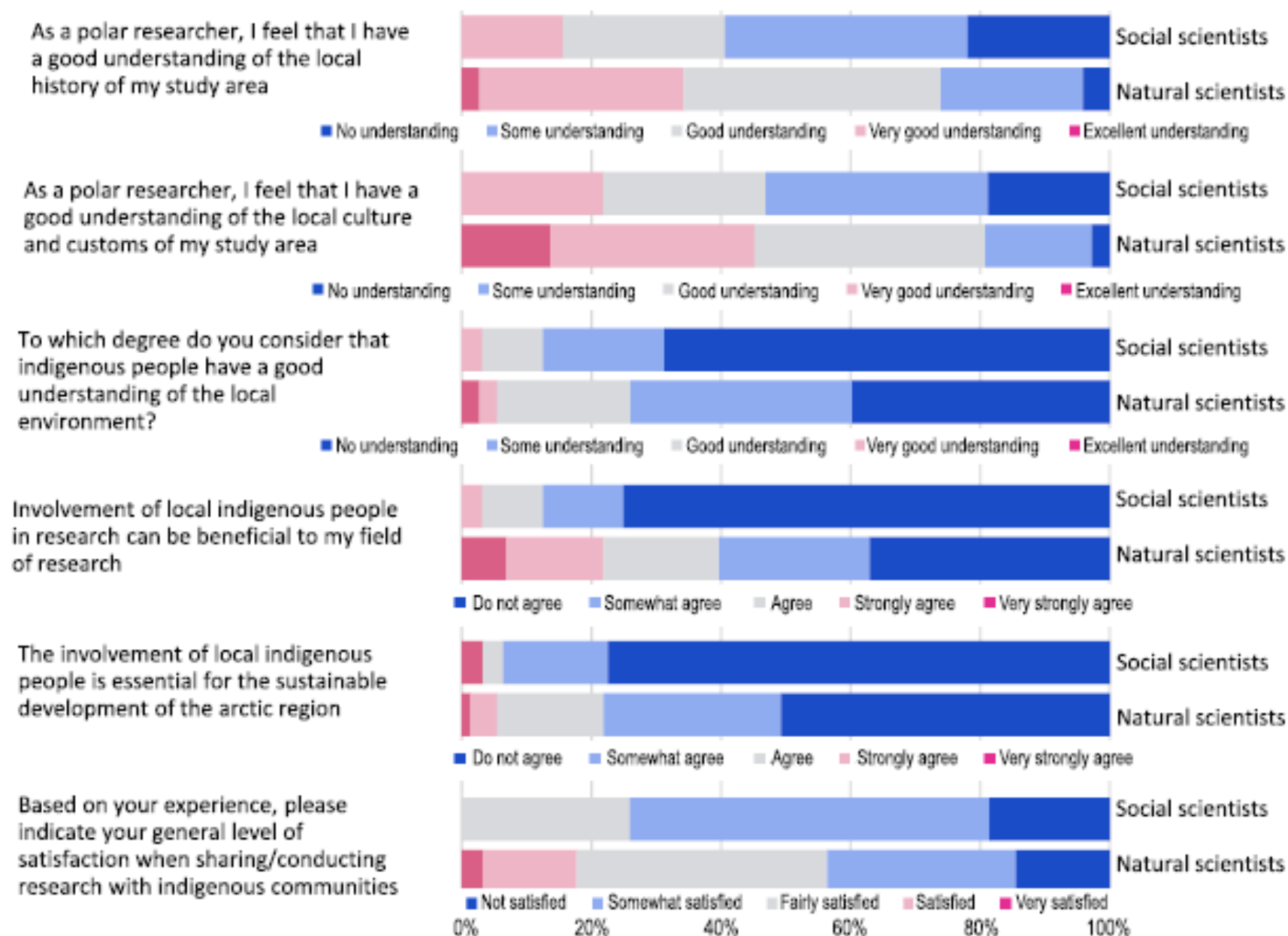


Fig. 6. Questions that were rated differently ($p < 0.05$) by social and natural scientists.



Sjöberg et al. (2015)

unequal knowledge distribution and power

how is the integration of Indigenous knowledge in the North American Arctic such a well-established practice but not in Greenland?

Home Rule vs homeland

- Greenland became 'independent' from Denmark in 1979
- but 'Nuuk remains as far away as Copenhagen' (Sejersen, 2004)

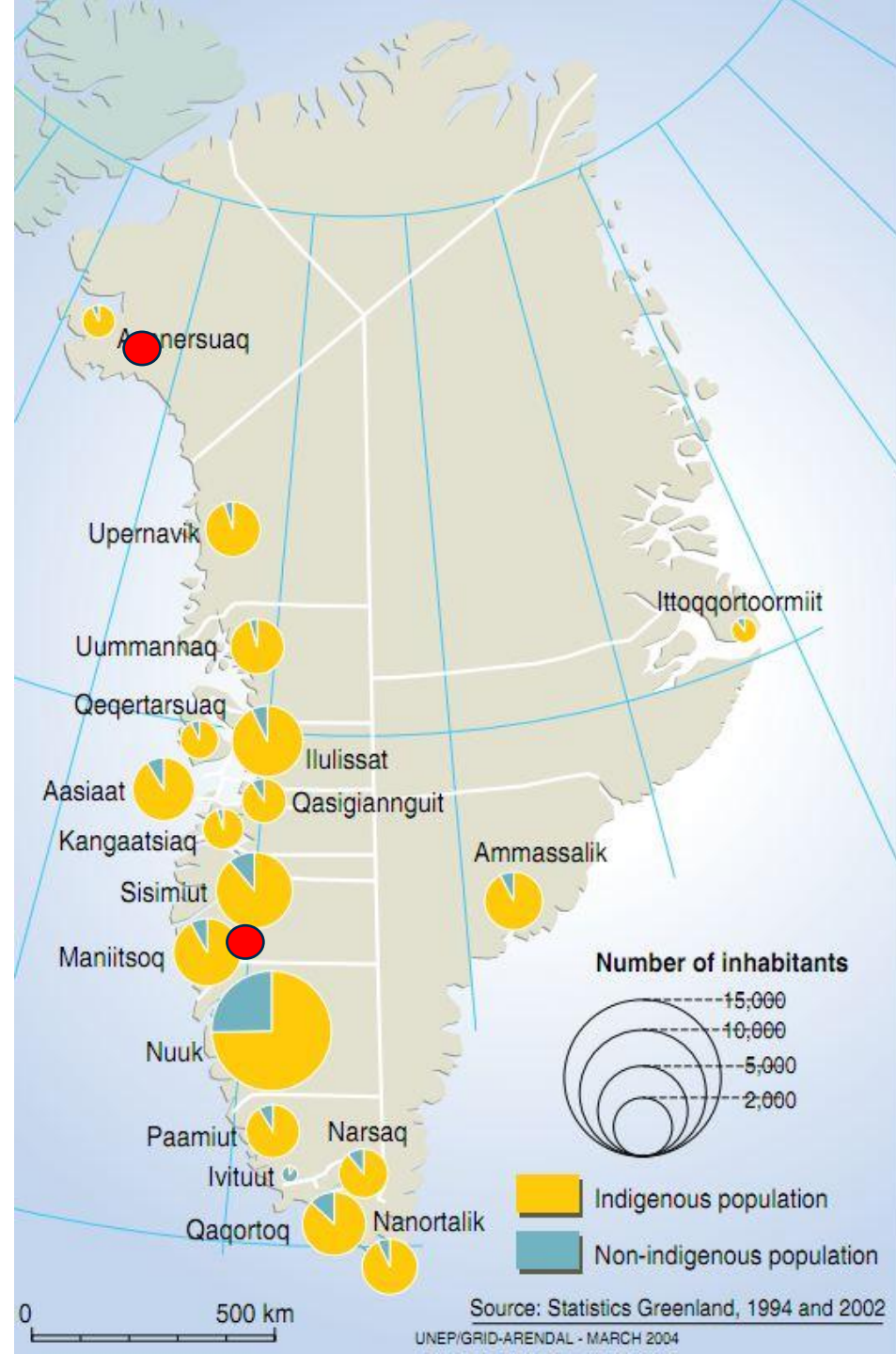
Field Work,
K Transsect, Greenland Ice Sheet
August 2019





0,036% of area
53% of population

99,64% of area
47% of population



historic legacy of gaining influence

North American Indigenous Societies were deprived of their homeland and had/have to regain control – an increased sense of ownership has grown that acts in a decentralized manner.

Greenland was never completely deprived of its land and is now in charge – through a centralized Home Rule that marginalizes people outside the capital. Local perspectives are voiced predominately through political institutions. *(Sejersen, 2004) – Local knowledge holders are now often equal to hunters, who naturally have a limited interest range not necessarily representative of all.*

- peripheral discontent with central authority -> see parallels to mountains

role of knowledge holders

North America – knowledge and political power is (was) held by Elders.

Greenland – past has a less positive connotation and Elders do not play a comparable role at the societal level. Focus is on maintaining a modern nation (Kalaallit Nunaat). *(Sejersen, 2004)*

role of knowledge holders



Eternity Fjord, Western Greenland
August 2018

role of knowledge holders



Eternity Fjord, Western Greenland
August 2018

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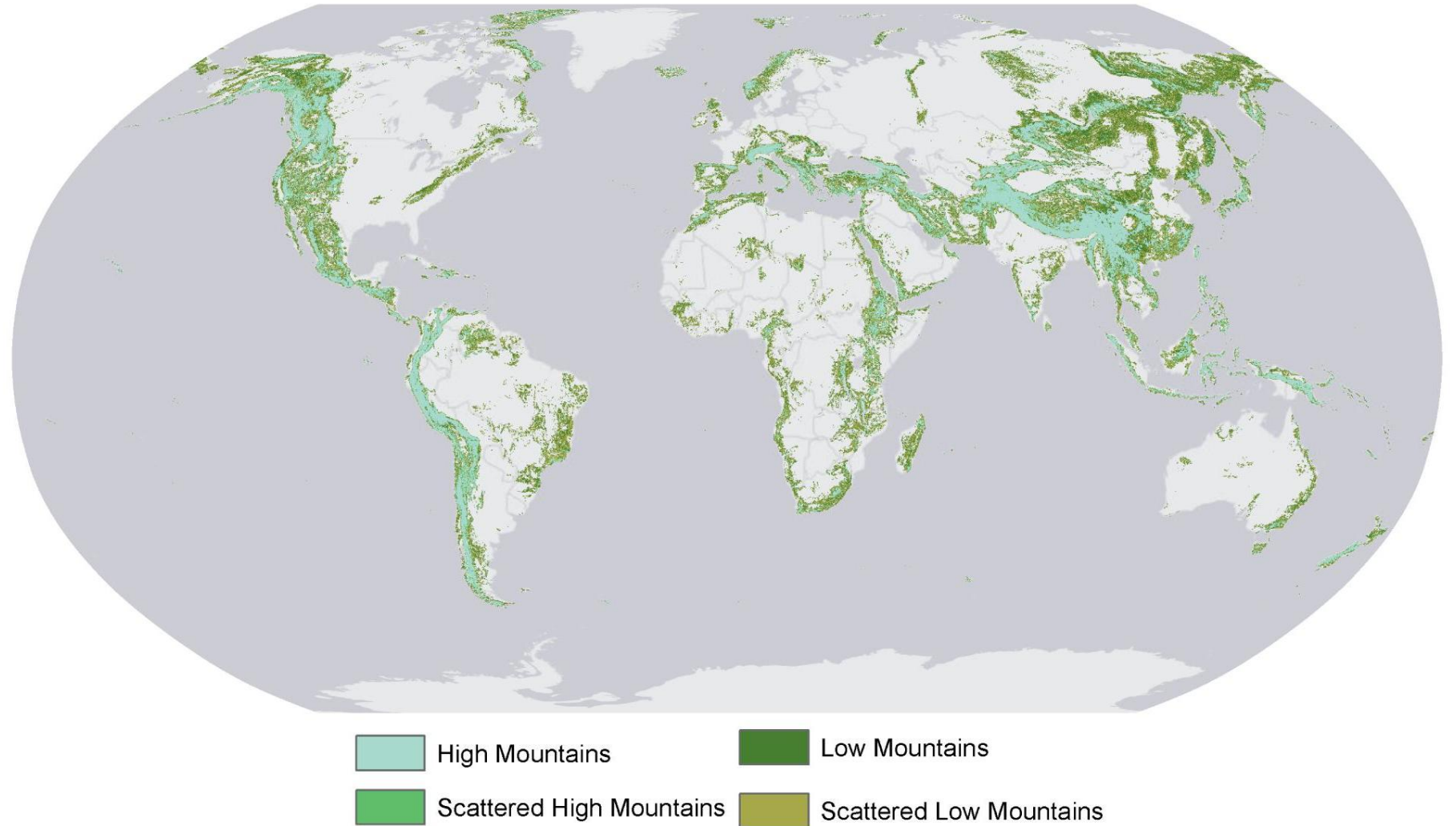
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ILK in mountains



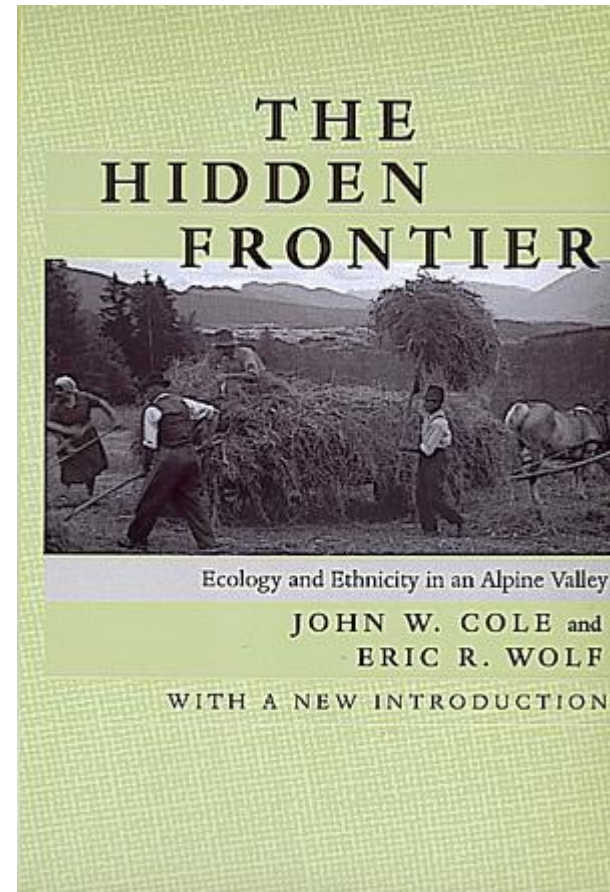
ILK in mountains

Large interest in

- 'remoteness', scattered populations – but 5 -10% of global population live in mountains
- challenge to maintain infrastructure, perceived 'disconnectedness'/'remoteness'
- ecology, natural hazards/disaster risk reduction (since the turn of the century), appreciating the 'vulnerability' (and 'coping') part of risk (hazard + exposure + vulnerability + ability to cope) (Blaikie et al, 1994)
- High Mountain Asia (HMA) and the Andes – why less on the Alps or the Apalachees? (see discussion on inclusion of ILK in the IPCC report)

ILK in mountains

- some anthropological work, especially with regard to remoteness and living on borders (see e.g. Cole and Wolf, 1974 – written by Americans about Europeans, investigating ‘the other’ – ‘othering’ - already present then)
- borderland studies (e.g. LMU Munich)
- 1980s anthropological work starts to compare Andes and the Himalaya (Orlove, 1987)



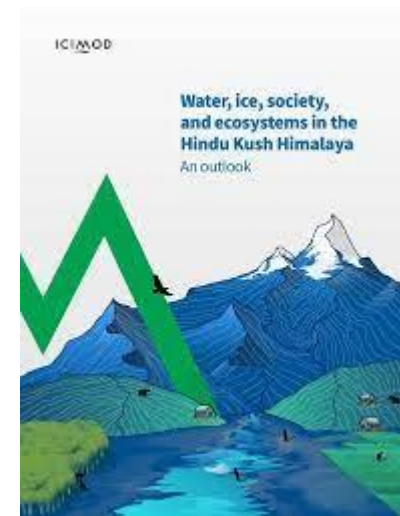
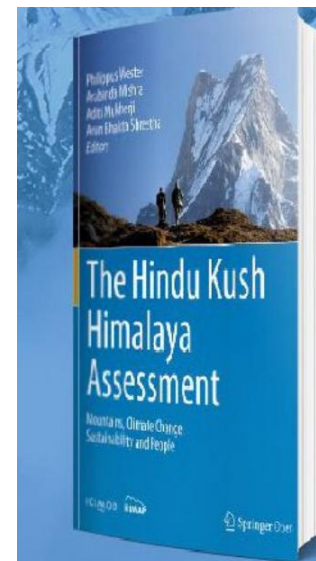
ILK and climate science in mountains

- explicit scientific literature on the topic starts after 2015
- appraisal in the IPCC AR6
- inclusion in studies as well as management scarce, limited to herding, medicinal plants and DRR

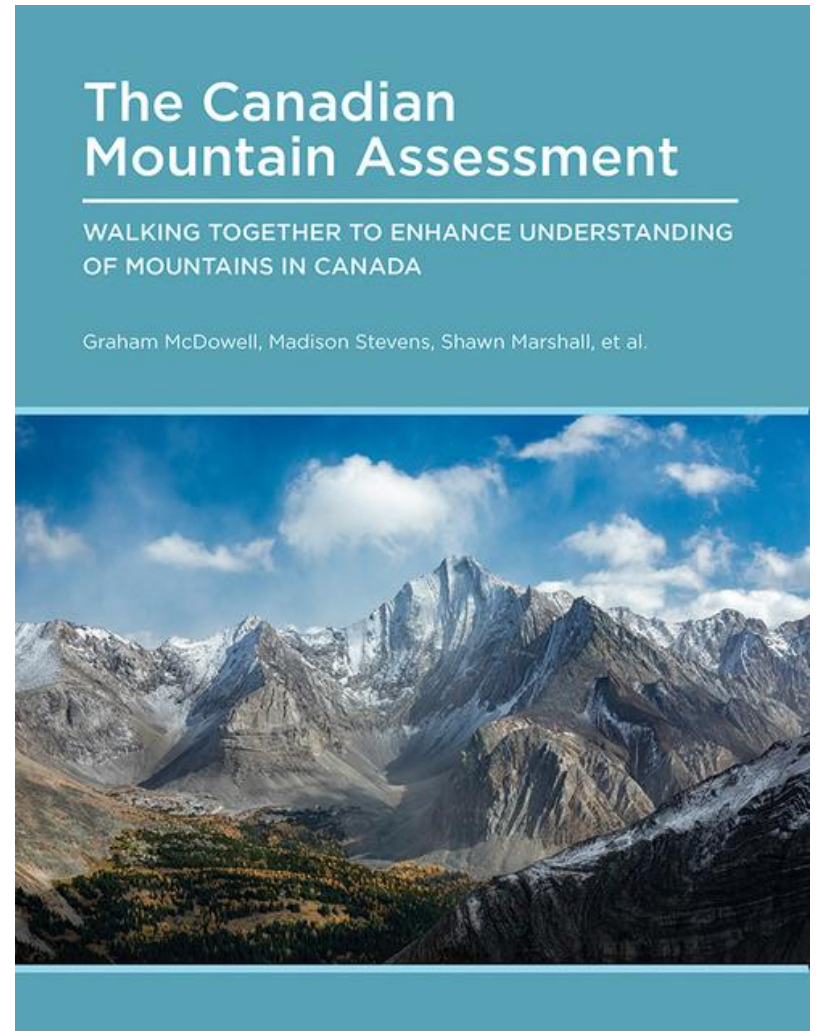
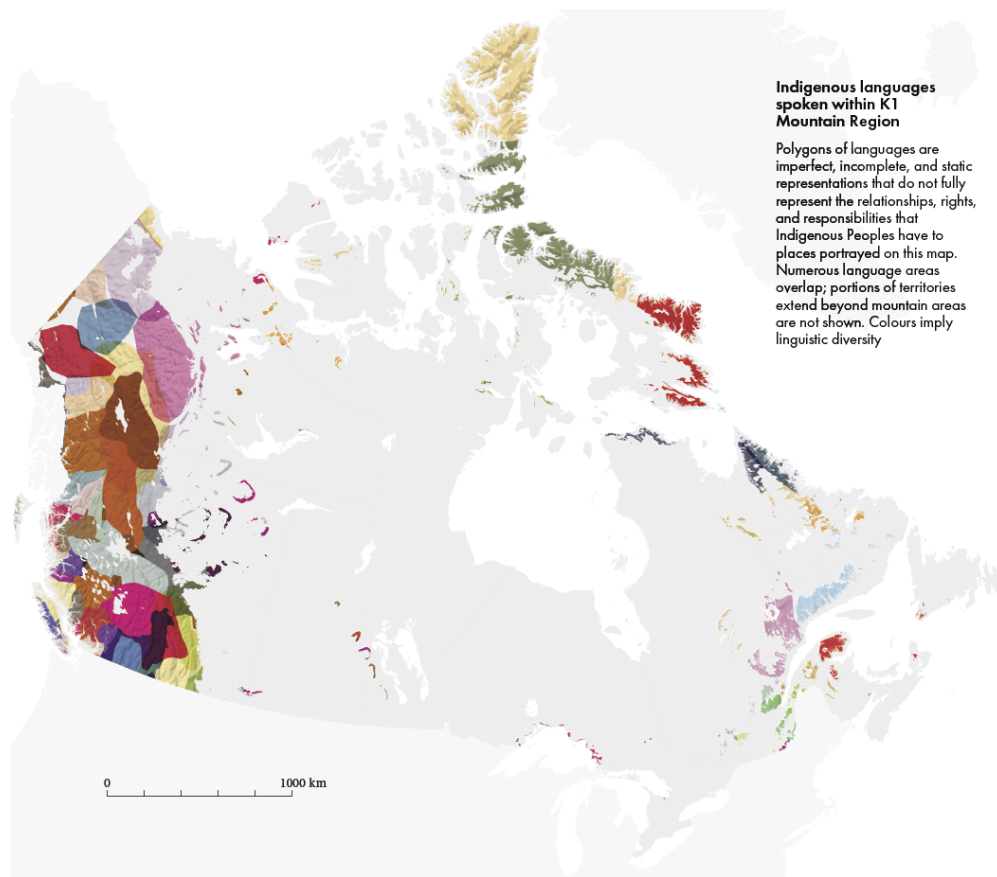
Example High Mountain Asia:

HiMAP Assessment 2019 – not covered

HiWISE Assessment 2023 – partially acknowledged



Example Canadian Mountain Assessment



<https://ucp.manifoldapp.org/projects/9781773855103>

Local knowledge in mountains

(1) Observation



People's experience of the local surroundings

History of natural hazards

Examples: knowledge on the location, time, duration, frequency, intensity, predictability of previous hazards

Nature of natural hazards

Examples: knowledge on the onset, origin, velocity of water flow; knowledge of different types of rain

Evolution of social and physical vulnerabilities to natural hazards

Example: life stories explaining the impoverishment processes of households following recurrent natural hazards and other stresses

(2) Anticipation



People's identification and monitoring of environmental indicators

Early warning signals

Examples: interpretational knowledge of changes in animal behaviours, vegetation patterns; knowledge of local weather forecast

Time thresholds

Examples: knowledge of when it is time to buy and store food in advance, leave the house, move the cattle, and remove important belongings

Escape route and safe places for humans and cattle

Examples: knowledge of the safest and fastest routes

Key actors and skills

Examples: knowledge of who knows what, who does what and when, who stays behind, who goes first

(3) Adaptation



People's access to assets and their ability to learn, self-organise, re-organise, innovate etc.

Human assets

Examples: specific skills such as traditional carpenters and masons

Sociocultural assets

Examples: knowledge of different social groups depending on occupational, physical ability, ethnicity, gender, caste, class, & age characteristics

Institutional assets

Examples: knowledge generated by local institutions and cross-scale linkages

Financial assets

Examples: micro-finance arrangements such as credits and savings

Natural assets

Examples: natural resource management strategies such as intercropping and agroforestry that conserve biodiversity and protect soil erosion and can contribute to reducing the impacts of natural hazards

Physical assets

Examples: infrastructural safety arrangements such as boats, housing, embankments

(4) Communication



People's ability to transfer knowledge among themselves and between generations

Oral & written communication

Examples: local songs, poems, proverbs which help the younger generation and outsiders to learn about previous hazards; stories of previous hazards encoded in the name of specific places

Early warning systems

Examples: use of visual signals such as mirrors, fire or audio signals such as drums; having dreams of natural hazards in advance

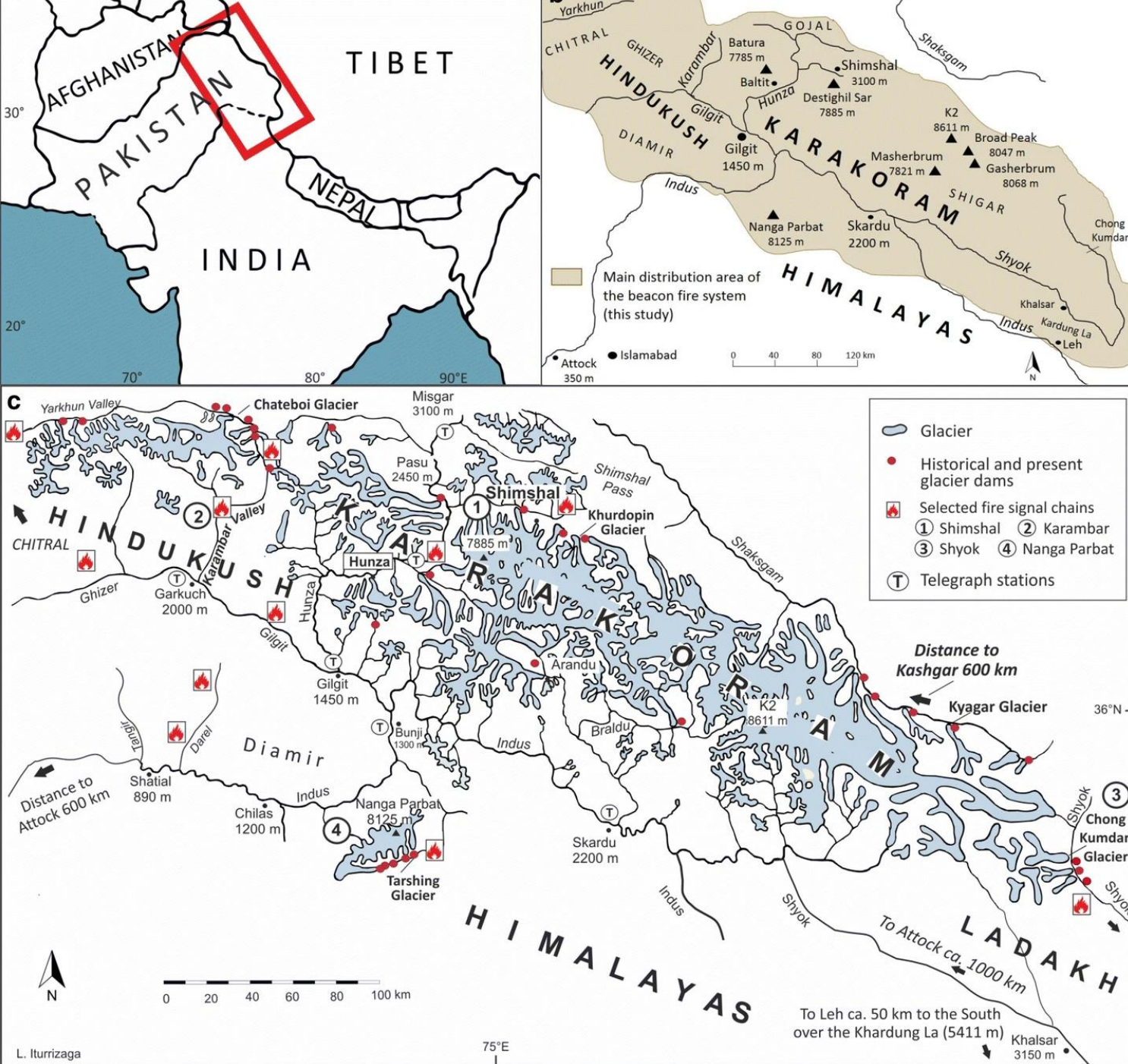
Other practices

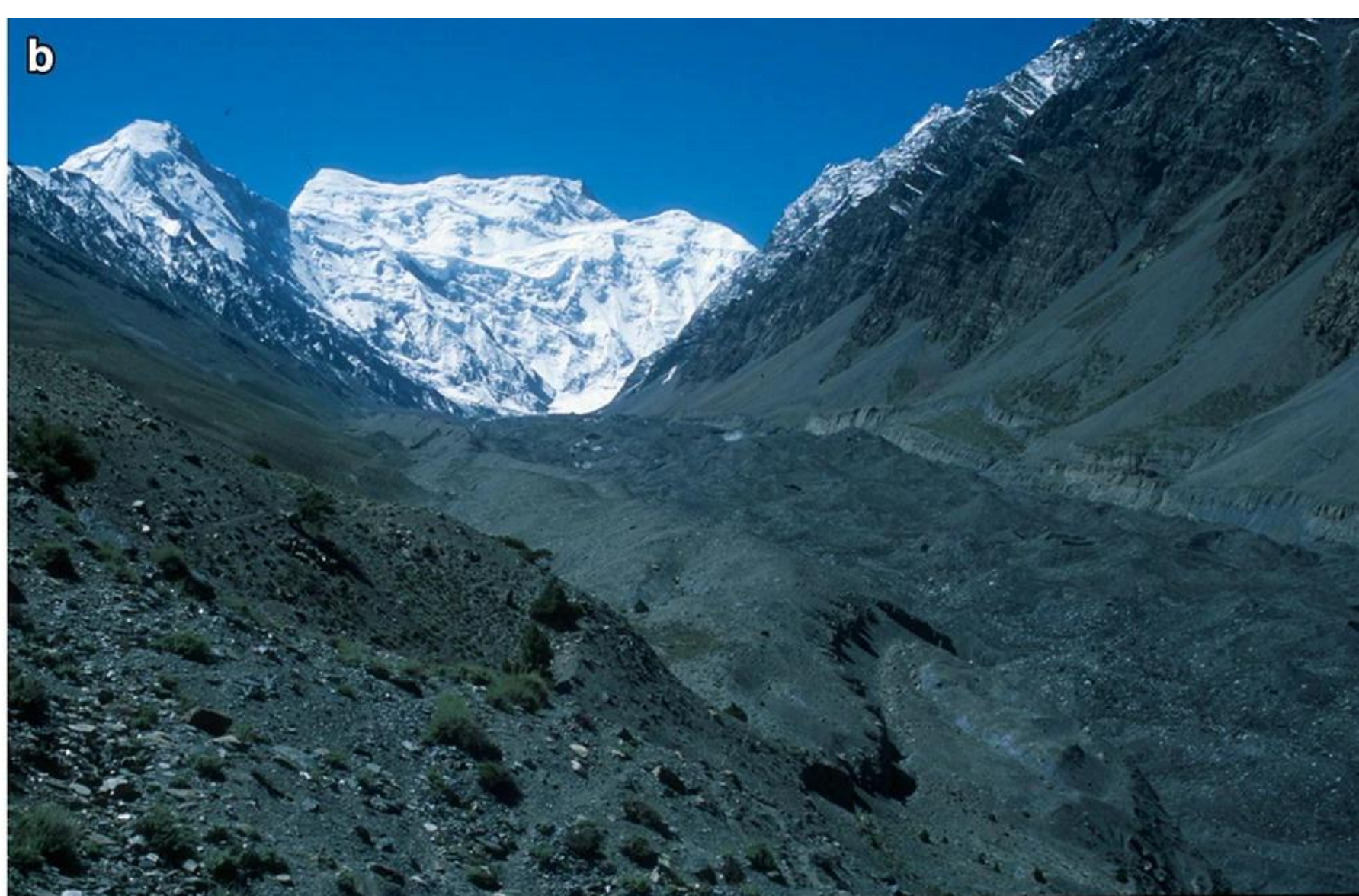
Examples: taboos which prevent people from going to certain hazard-prone areas; ceremonies, local art which helps the community to understand, remember past natural hazards, and relieve the anxiety related to the threat of future hazards

(Lost?) communication skills

Early Warning of mountain
floods with beacon fires

Iturrizaga 2007





Glacier growing, hazards and agriculture

- collect (male and female) glacier ice
- store it in high caves (permafrost) where more snow grows it
- access in spring/dry years for irrigation



Tveiten 2007

Glacier growing, hazards and agriculture

A



B



C

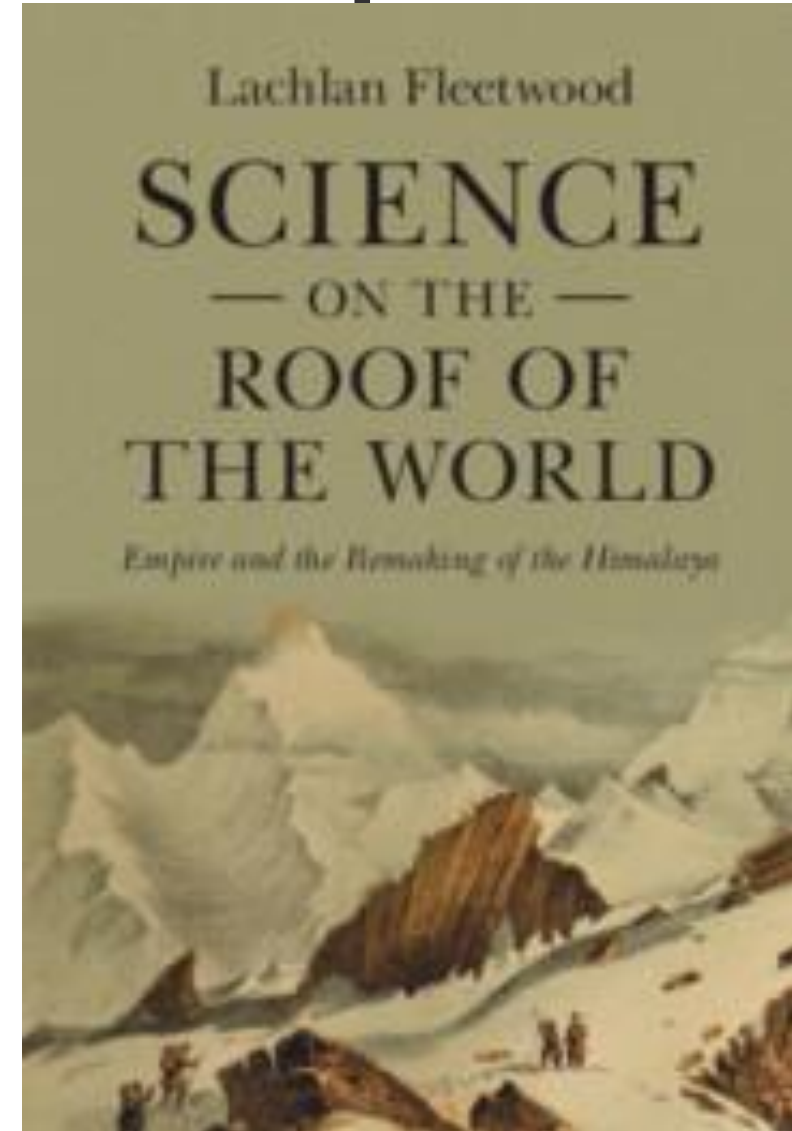
unequal knowledge distribution and power

'local' knowledge centers in High Mountain Asia have in the past as well as in the present often been far away from the actual mountains.

Is someone from Calcutta 'local' to the Indian Himalaya?

Is someone from Kathmandu 'local' to the Everest region?

<https://publicanthropologist.cmi.no/2024/06/06/fragile-instruments-and-himalayan-frontier-making-science-in-the-himalayas/>



invisibility of knowledge

- a lot of knowledge around (non-technical) practices is 'invisible' (not documented, taken for granted)
- knowledge of marginalized groups (e.g. women) can remain hidden (or at least not accessible to outsiders)



Community Interaction,
Melamchi, Nepal
May 2022

invisibility of knowledge

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- knowledge of marginalized groups (e.g. women) can remain hidden (or at least not accessible to outsiders)
- knowledge not in the spotlight (e.g. related to 'slow onset' hazards) remain out of view

Community Interaction,
Shigar, Pakistan
July 2023



Drought
Khaplu, Pakistan
July 2023



Drought
Khaplu, Pakistan
July 2023



traditional knowledge and maladaptation

What to adapt to if the old rules do not apply anymore? -> rapid climate change

At what time scale can ILK contain knowledge on a changing environment?

Literature

Acharya et al. (2023): Review article: Snow and ice avalanches in high mountain Asia – scientific, local and indigenous knowledge, <https://doi.org/10.5194/nhess-23-2569-2023>

Agrawal, A. (1995): Dismantling the Divide Between Indigenous and Scientific Knowledge, Development and Change doi:10.1111/j.1467-7660.1995.tb00560.x

Blaikie, P. et al. (1994): At Risk - Natural Hazards, People's Vulnerability and Disasters, <https://doi.org/10.4324/9780203974575>

Cole and Wolf (1974): The Hidden Frontier Ecology and Ethnicity in an Alpine Valley, UCPress

Dekens, J. (2007): Local Knowledge for Disaster Preparedness: A Literature Review, ICIMOD

Orlove, B.S. (1987) Mountain anthropology and mountain anthropologists: The comparative study of populations at high elevations, Reviews in Anthropology, 14:2, 95-100, DOI: 10.1080/00988157.1987.9977812

Sejersen, F (2004), Local knowledge in Greenland - Arctic perspectives and contextual differences. in DG Anderson & M Nuttall (eds), Cultivating Arctic Landscapes: Knowing & Managing Animals in the Circumpolar North. Berghahn Books, Oxford, pp. 33-56.

Shrestha et al. (2023): A comprehensive and version-controlled database of glacial lake outburst floods in High Mountain Asia, <https://doi.org/10.5194/essd-15-3941-2023>

Sjöberg et al. (2018): Involvement of local Indigenous peoples in Arctic research — expectations, needs and challenges perceived by early career researchers, <https://doi.org/10.1139/as-2017-0045>

Smith, H. A., Sharp, K. (2012): Indigenous climate knowledges, WIREs Climate Change <https://doi.org/10.1002/wcc.185>

Tveiten (2007): Glacier Growing - A Local Response to Water Scarcity in Baltistan and Gilgit, Pakistan, http://www.umb.no/statisk/noragric/publications/master/2007_ingvar_tveiten.pdf

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Disaster Risk Reduction (DRR)

Disaster Risk Reduction is the concept and practice of reducing disaster risks through systematic efforts to analyse and manage the causal factors of disasters (UNDRR)

Loss and damage caused by climate change has already uprooted the lives and livelihoods of millions of people, and extreme weather events are endangering communities everywhere. One-third of Pakistan has been submerged. Pacific island nations are on the brink of extinction due to rising water levels.

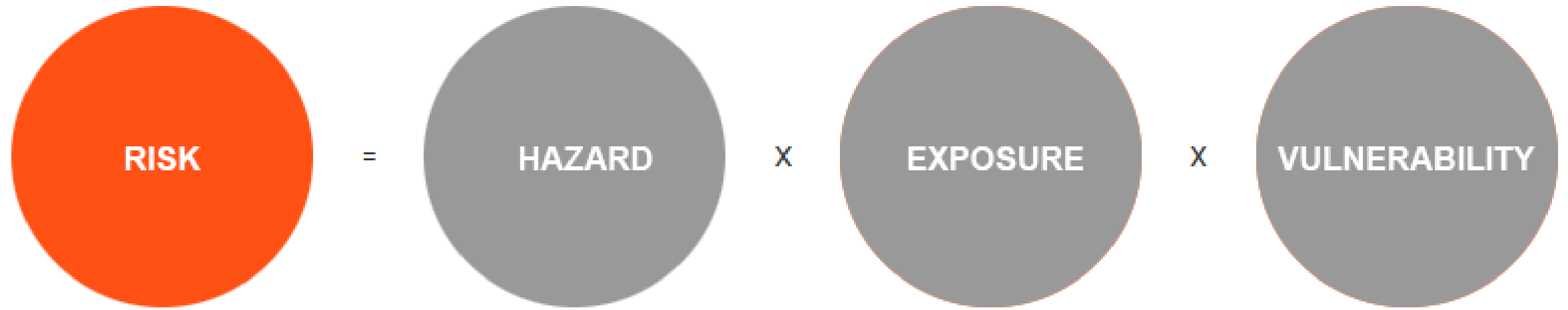
In 2020 alone, 980 natural disasters cost the global economy more than US\$210 billion.

The future outlook is even more serious.

Without action on climate risk reduction, it is projected that by 2030 there will be more than one disaster every day.

According to a recent UNDP study, if emissions reductions remain moderate, climate change could kill more than 40 million people by the end of the century.

Asako Okai, ASG UNDP, March 2023 - <https://www.undp.org/speeches/directions-disaster-risk-reduction-era-climate-change>



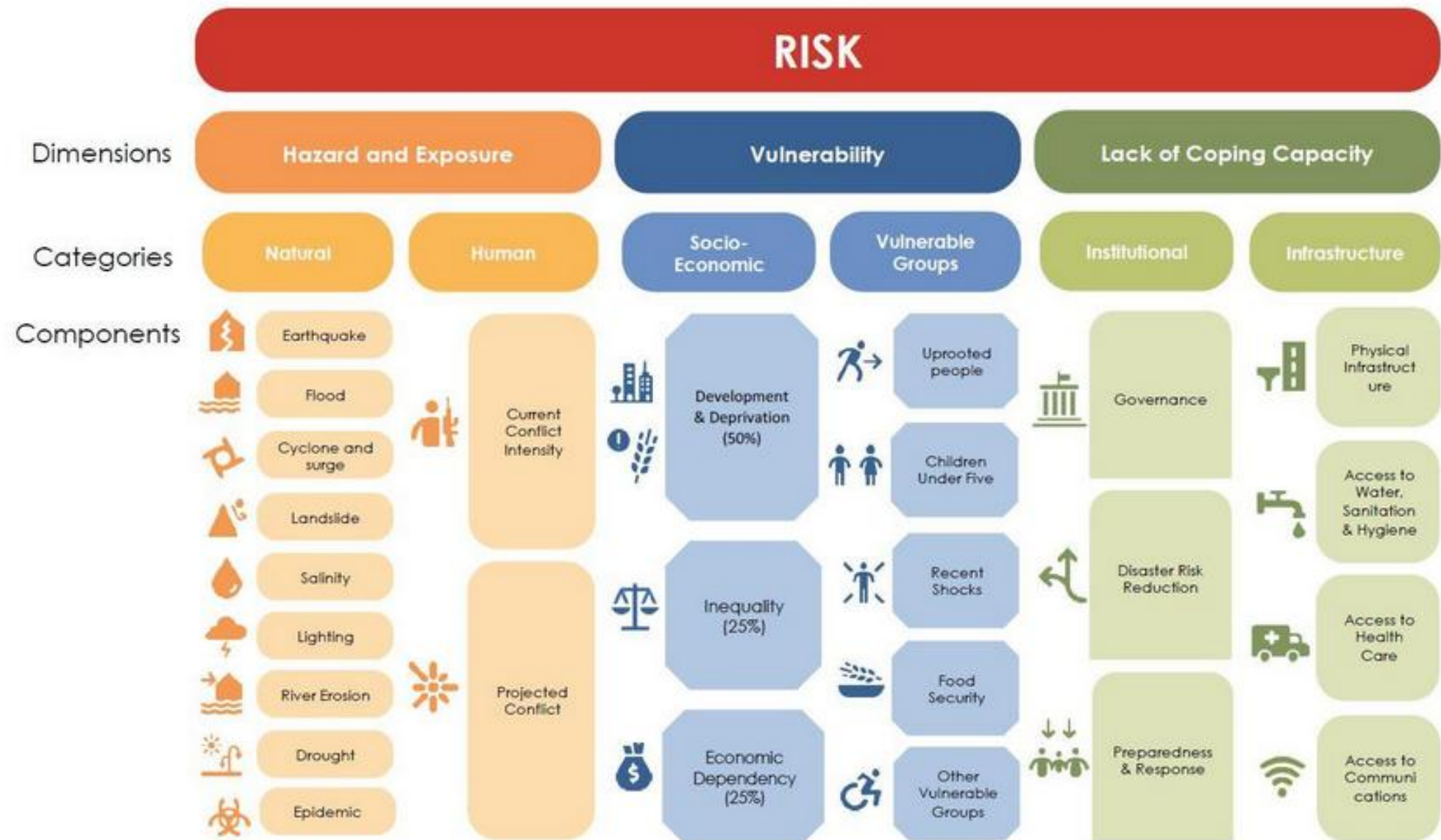


Illustration: Risk dimensions, categories and components retrieved from [Bangladesh INFORM Sub-National Risk Index 2022](#), UNDRR (2022)

DRR and climate change

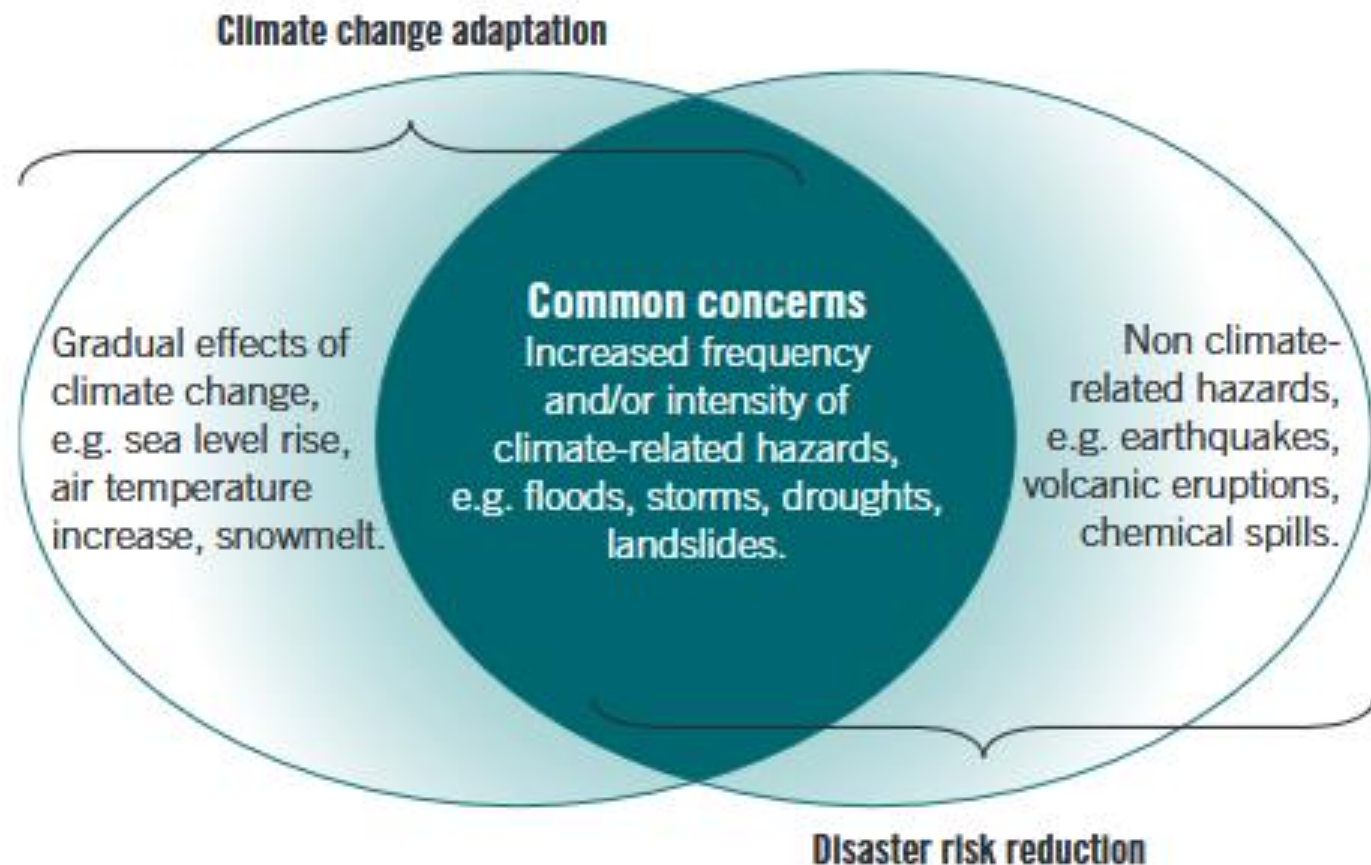


Figure 1.1: Common concerns of climate change adaptation and disaster risk reduction

Role of ILK in DRR

- Long established technical know how and practices to be prepared for disasters
- What could go wrong with climate change?
- What could go wrong with socio-economic change?

Example of local knowledge and earthquakes

kath-kuni
architecture in
the Himalaya



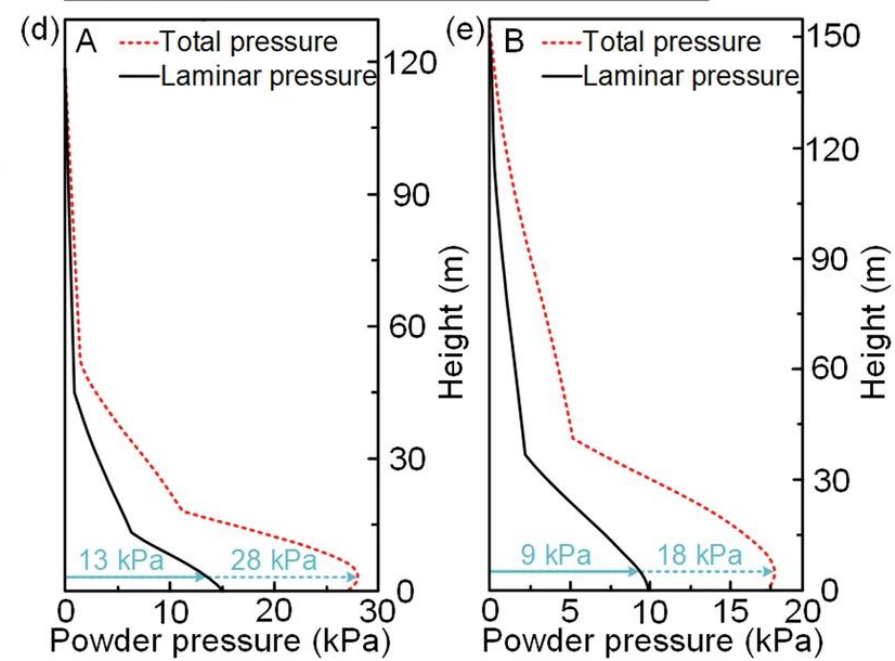
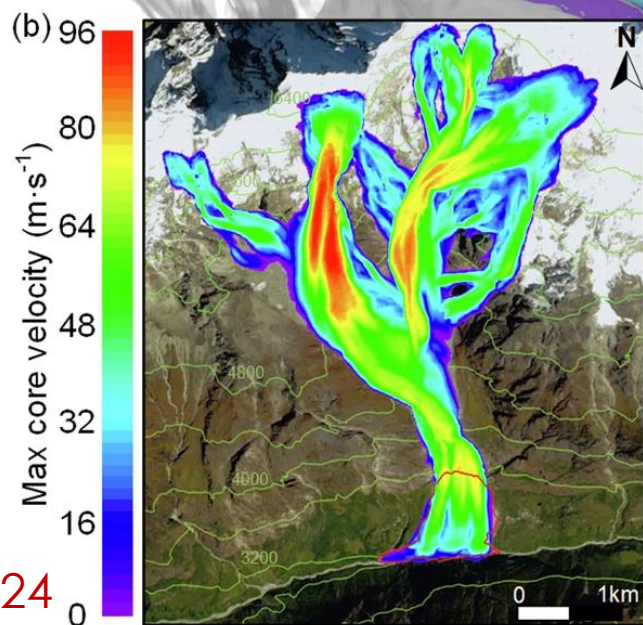
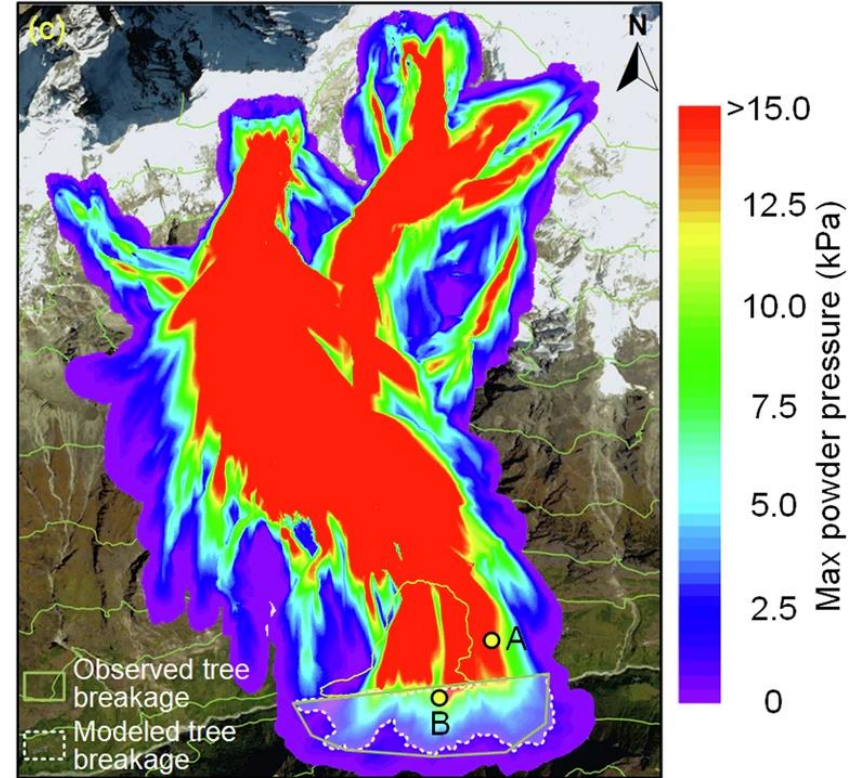
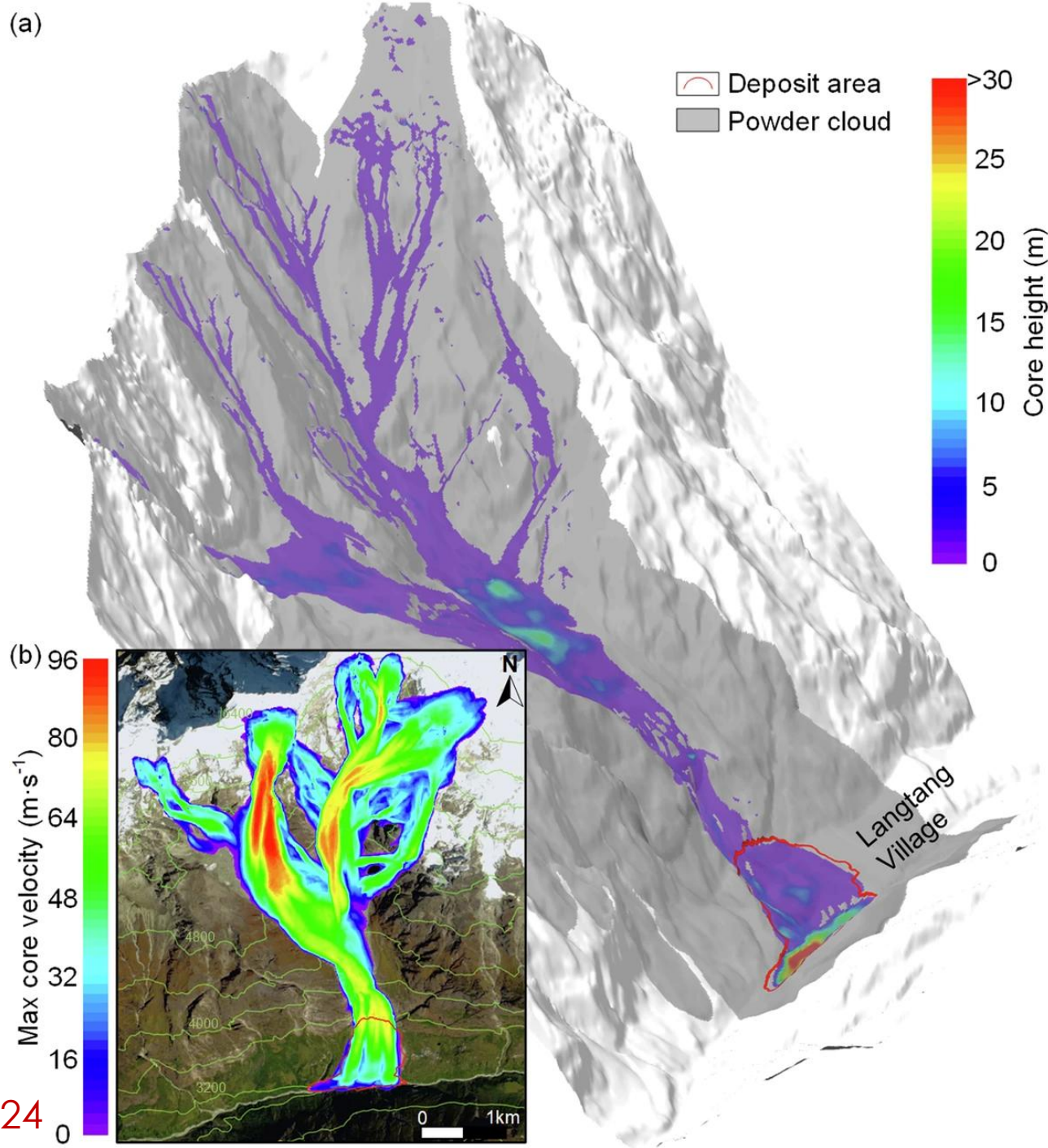
<https://www.bbc.com/travel/article/20220825-kath-kuni-the-himalayas-ancient-earthquake-defying-design>

Example of local knowledge and earthquakes

Traditional main rooms
in the Karakoram-
Hindukush-Pamir



(a)





ILK and DRR

Explore the resources provided [here](#) and discuss – are there examples from your own background that are documented like this? Does such documentation exist for the 2021 floods in Central Europe? If yes, is it any different? If you can't find anything similar, why does this not exist?

<https://www.undrr.org/words-action-using-traditional-and-indigenous-knowledges-disaster-risk-reduction>

Read case studies 1-12 (pages 71-95). After reading any of the case studies, reflect on any one of the 4 questions shown below. You will have an opportunity to discuss your answers with students in class too.

- (1) Explain how hazards intensity/frequency can be linked to the exclusion of indigenous peoples from their land?
- (2) What are some of the obstacles to adaptation (to hazards) Indigenous communities face?
- (3) What are some of the drivers of indigenous and local knowledge loss? How does this loss contribute to increasing disaster risk?
- (4) Are there any examples from your own community that are like the case studies documented in this report? If yes, what are some similarities and differences between your own example and those included in this report? If no, why is that so? For example, consider the 2021 floods in central Europe. Can you find any documentation of Indigenous or local knowledge in this context?

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Integrating ILK and science

Science as in 'conventional' or 'Western'

Initial considerations

- at the basis should be autonomy for Indigenous/local or traditional knowledge holders (Agrawal, 1995)
- IL knowledge holders need to be integrated in the process (rather than just 'taking data from them' -> ethics) and whatever outcomes are generated need to be shared with communities
- data types not mutually exclusive
- data can be complementary

ILK and science in HMA – example GLOFs

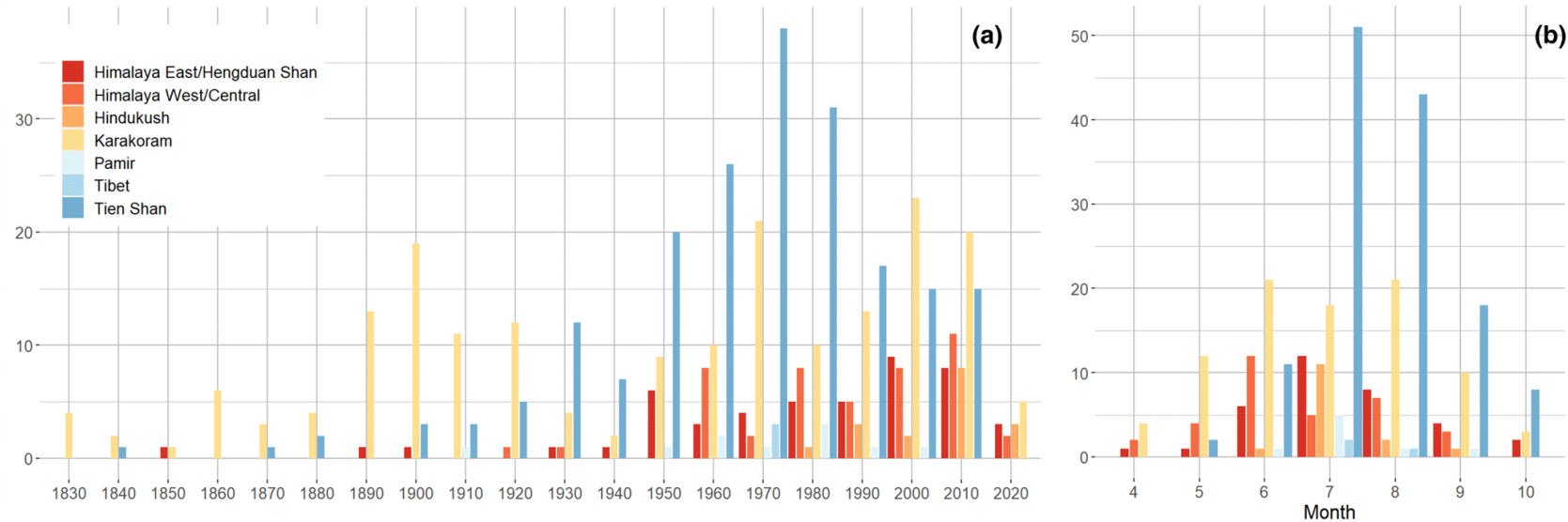


Figure 3 (a) Temporal evolution of recorded GLOF occurrence per region and decade. (b) Seasonal occurrence of GLOFs. Months November to March are not shown, with total events < 5.

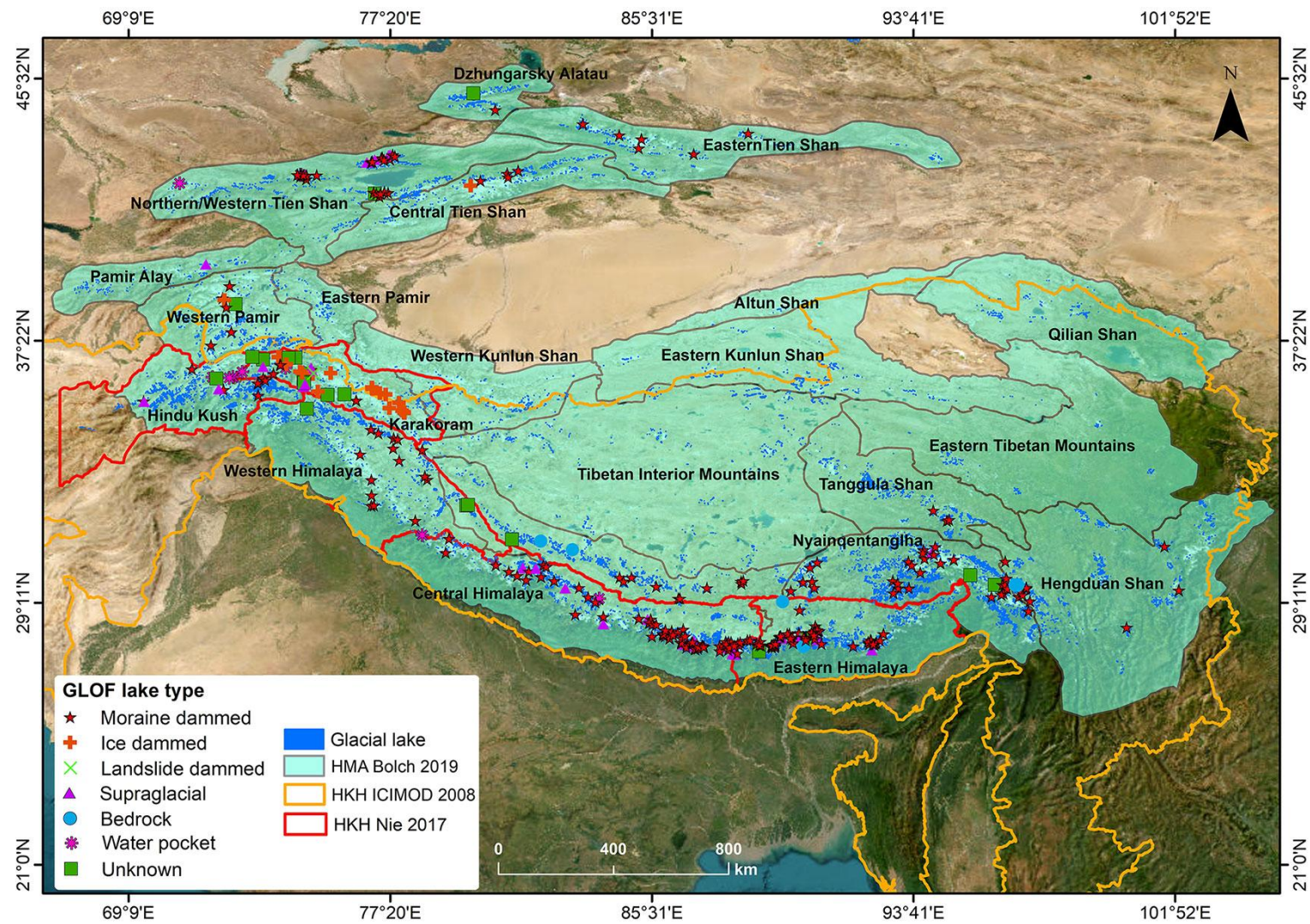
Shrestha et al. 2023

- GLOFs happen often without gaining international attention
- even if not killing people can affect society (land, heritage, psychology)
- many have happened while 'satellites weren't looking'

Example: ILK and satellite imagery

Who gets to decide if an event happened? Who is 'more trustworthy'?

Satellites or local people?



ILK and science in HMA – example GLOFs

Observed trends and progress

- Human dimensions are generally underrepresented in GLOF papers covered by WOS and Scopus databases, partially reflecting the publishing paradigm of individual disciplines (for instance, a tendency towards publishing books in the social science domain)
- Recently increased focus on GLOF attribution to (anthropogenic) climate change (see Sect. 4.7), with natural scientists generally driving this research and only at times involving social science or humanities researchers (Harrison et al., 2018; Huggel et al., 2020a; Stuart-Smith et al., 2021)

Identified challenges

- Better consideration and appreciation of cultural and spiritual values of glacier lakes and high-mountain environments and local and indigenous knowledge in GLOF research (Lambert and Scott, 2019; Abdel-Fattah et al., 2021; Matti and Ögmundardóttir, 2021; Haerberli and Drenkhan, 2022)
- More sustained, detailed work needed to understand diverse and intersecting drivers of risk and vulnerabilities in communities (Matti et al., 2022b; see Sect. 4.6) to reveal how societal variables making diverse populations vulnerable overall intertwine with GLOF risks (Carey et al., 2020)
- The politics and cultures of GLOF risk management and responses are only partially understood and analysed (Moulton et al., 2021)
- Linking GLOF occurrence patterns with anthropogenic climate change signal (Roe et al., 2021; see Sect. 4.7)

Proposed ways forward

- Recognition and empowerment of (often powerful) local communities and indigenous knowledge (Mercer et al., 2010; Kelman et al., 2012; Carey et al., 2021; see Sect. 4.6); avoid research colonialism (come, do research and leave) and involve local researchers
 - Employment of integrated, holistic, interdisciplinary approaches (Carey et al., 2012; Gall et al., 2015) and consideration of overlooked dimensions and aspects (e.g. spiritual and cultural), knowledge co-production and exchange
 - Development of robust methods (statistical methods and process-based models) for attributing GLOF occurrence and impacts to anthropogenic climate change (Stuart-Smith et al., 2021)
-

ILK and science in HMA – example GLOFs

Crucial downside with local knowledge on GLOFs

- it has been compromised by Western research and development work on the topic. GLOFs get global attention so there is a motivation to call anything a GLOF.
- it could damage tourism potential, so you downplay their existence

What we lack so far is (meta-)research on how science and ILK influences (biases?) each other.

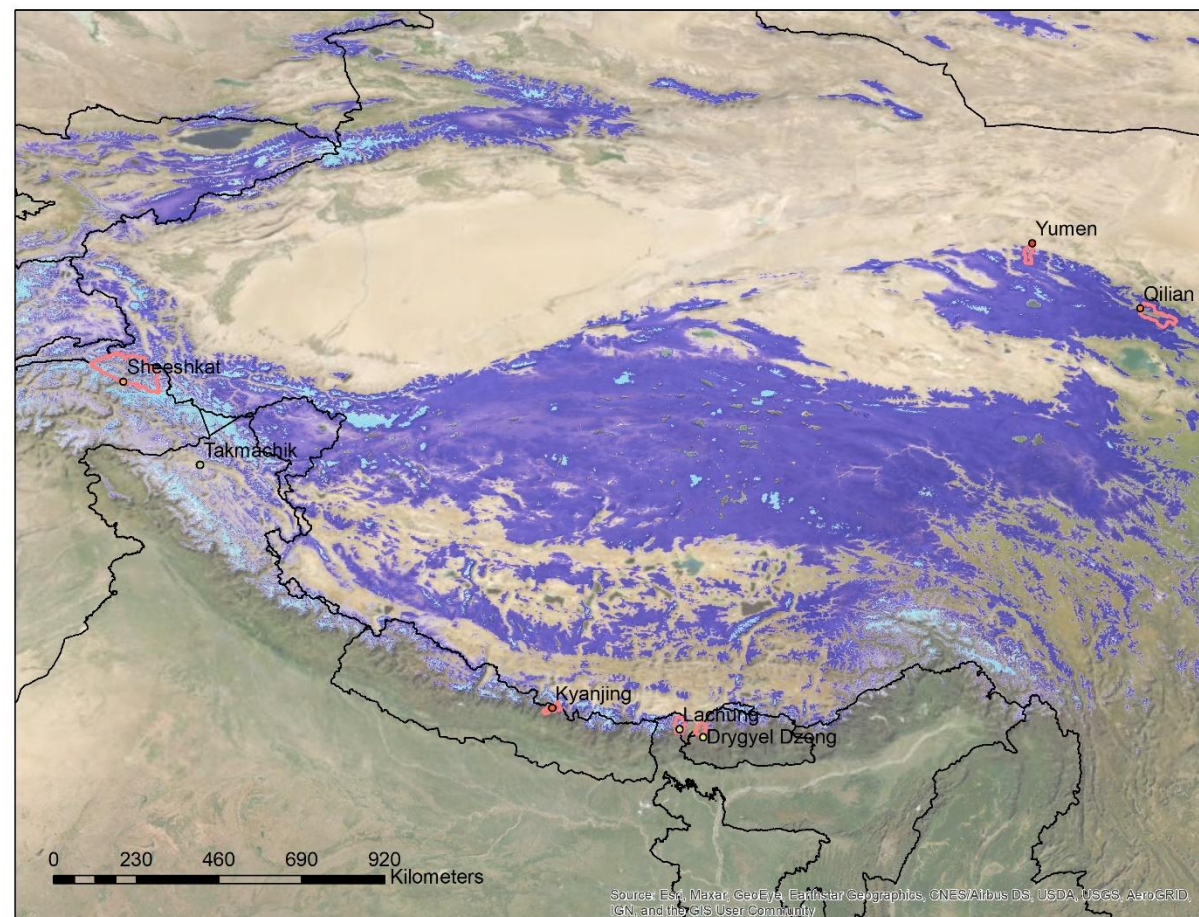
ILK and science in HMA – example climate

Ongoing study on cryosphere and society

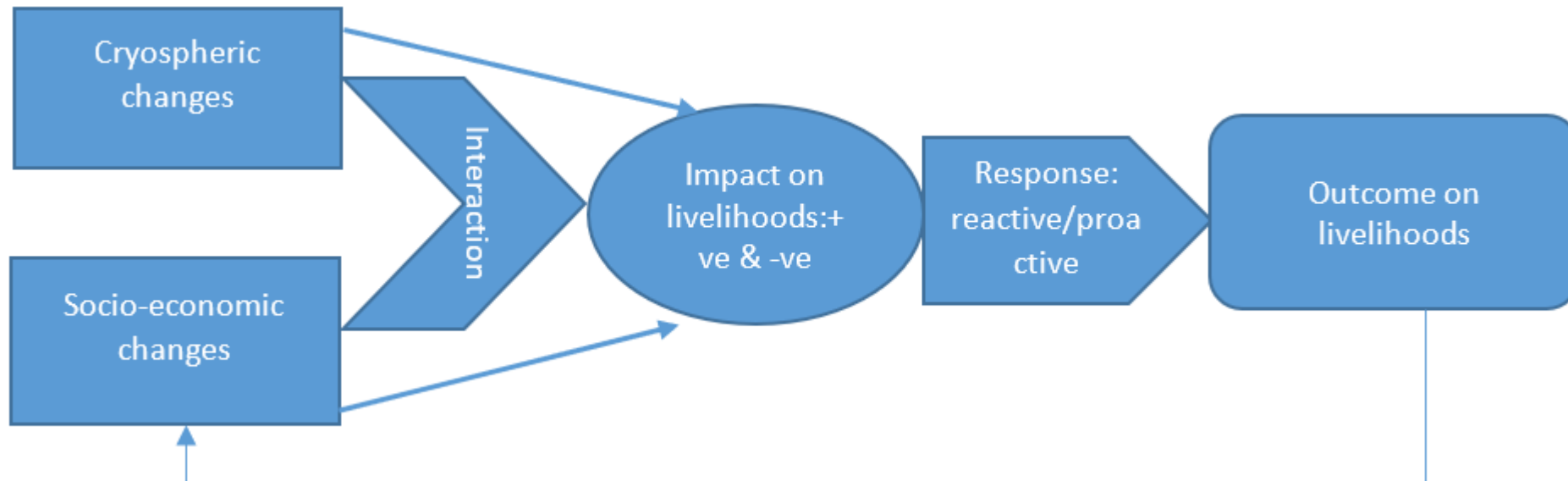
How are people impacted by snow, ice, permafrost and associated changes?

Do the perceptions match with what we see in conventional data?

Is 'conventional' cryosphere research relevant for communities 'supposedly' affected?



ILK and science in HMA – example climate



ILK and science in HMA – example climate

Table 2: Details of methods of data collection at each site

Methods of data collection	Meitshi-Shana, Bhutan	Yumen, China	Takmachik, (Ladakh) India	Lachen, (Sikkim), India	Langtang, Nepal	Shishkat, (GB), Pakistan
Group discussions	2	-	3	2	3	4
Key Informant Interviews	28	22	24	21	25	21
Field Observation	√	√	√	√	√	√
Household Survey	-	-	-	-	108 households	-
Cropping calendar	√		√		√	√
Historical timeline of hazards	√	√	√	√	√	√
Livelihood calendar	√				√	√

Tuladhar et al. in preparation

ILK and science in HMA – example climate

Hydrological Components		Direction of Change					
General component	Specific variable	Meitshi-Shana, Bhutan	Yumen, China	Takmachik, (Ladakh), India	Lachen (Sikkim), India	Langtang, Nepal	Shiskat, (GB), Pakistan
Cryosphere	Glacier Area	↓	↓	- NA	↓	↓	↑
	Glacier volume (thickness)	↓	↓	- NA	↓	↓	↓
	Snow cover area (SCA)	↓	- NA	↓	↓	↓	↓
	Snow volume thickness	↓	- NA	↓	- NA	↓	↓
	Permafrost	- NA	↓	- NA	- NA	- NA	- NA
	Aufeis	↓	- NA	- NA	- NA	↓	- NA
Precipitation	Snowfall amount	↓	↓	↓	↓	↑	↓
	Snowfall variability (seasonality)	↑	↑	↑	↑	↑	↑
	Rainfall amount	↑	↓	↑	↔	↑	↑
	Rainfall variability	↑	↑	↑	↑	↑	↑
	Transition from snow to rain	- NA	Yes	Yes	Yes	↑	- NA
	Transition from snow to hail	Yes	- NA	- NA	Yes		- NA
Discharge	Discharge amount in snow and glacier-fed rivers	↑	↑	↓	↓	↑	↑
	Discharge amount in non-snow and glacier-fed rivers	↑	- NA	- NA	- NA		- NA
	Discharge spring	↓	↑	↓	- NA		↔

ILK and science in HMA – example climate

Climatic Hazards	Meitshi-Shana, Bhutan	Yumen, China	Takmachik, (Ladakh), India	Lachan (Sikkim), India	Langtang, Nepal	Shiskat, (GB), Pakistan	Meitshi-Shana, Bhutan	
	GLOF	- N/A	- N/A	↔ ↑	- N/A	↑ N/A	1960s, resettlement of a village	
	Avalanche	↑	- N/A	- ↑	↑	↑ N/A	No incidences	
	Flood	↑ N/A	↑ ↓	↑	- ↑	↑ Flash flood in 2010	Frequent during monsoon, increasing	
	Landslide	↑	- ↑	↑	↑	↑ N/A	Frequent during monsoon	
	Windstorm		↑ 2010	↑ ↓ (Strong winds are a natural climatic phenomena and not a hazard)	- NA	(Strong winds are a natural climatic phenomenon and not a hazard) ↓		
	Cloudburst	↑ N/A	N/A	- N/A	↑	↑ 2010	N/A	
	Temperature (average)	↑	↑	↑	↑	↑	↑	

Tuladhar et al. in preparation

ILK and science in HMA – example climate

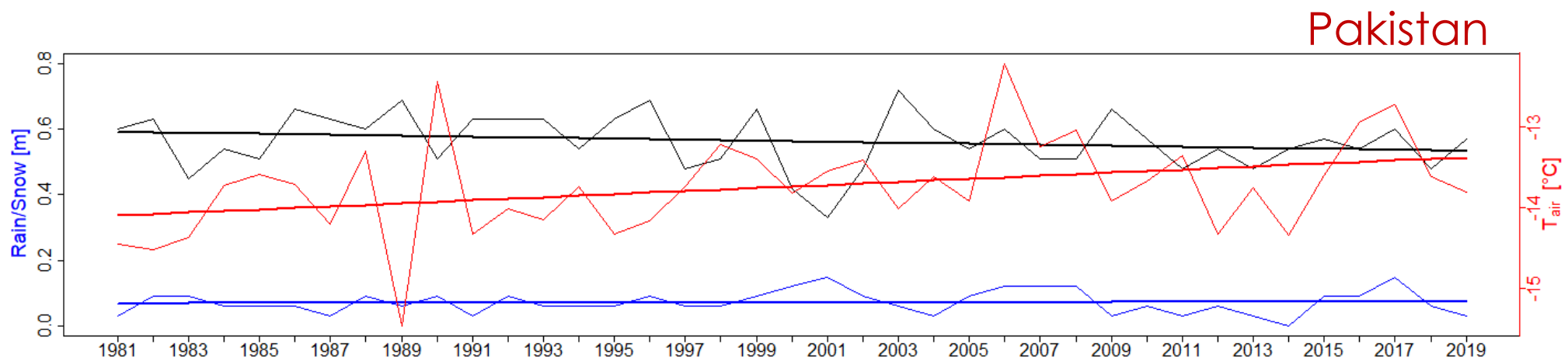
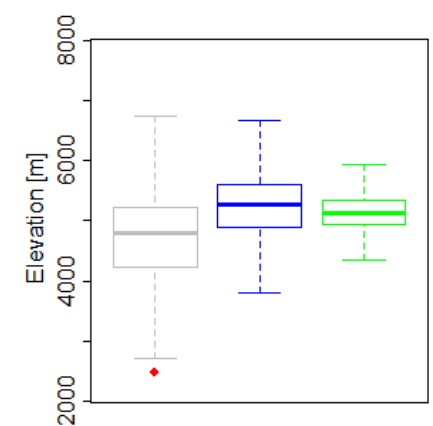
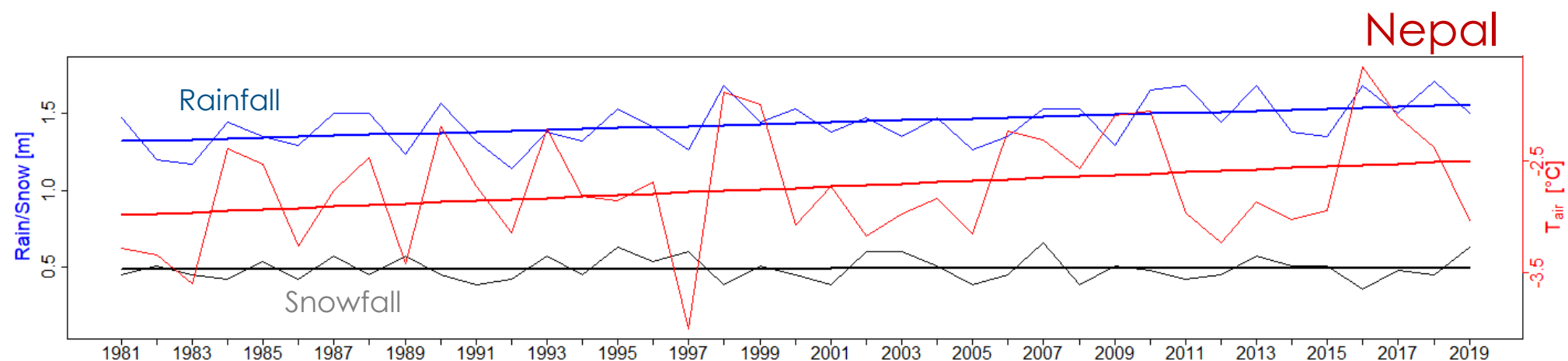
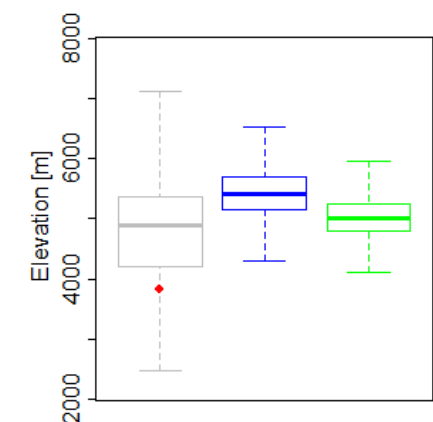
“If there is low snowfall early in the year, we can expect untimely heavy rainfall later in the year, as they are somehow linked, but we don't know how.” (female, Pakistan).

“About 15-20 years ago, the mountains around our town used to be covered with snow from October end, but nowadays mountain remain barren in the same month. We cannot predict snowfall times anymore as we used to before” (male, Bhutan).

"intensity of avalanche mainly depends on the intensity of snowfall in the winter season and I observe that such events tends to decrease with the passage of time owing to decrease in the intensity of snowfall on the mountains of the village. “ (m, Pakistan)

“Snow and glaciers are sources of life for us. Snow and rains are a blessing from the sky.”

ILK and science in HMA – example climate



Total area
Glacier covered area
Permafrost area





International efforts to integrate ILK

Table 1. Assessment of Respectful and Ethical Inclusion of TEK in the UNFCCC, the Kyoto Protocol, the AR4, and the ACIA

Indicator	UNFCCC	Kyoto Protocol	AR4	ACIA
Terminology consistent with indigenous peoples practices	No	No	Select chapters	Select chapters
Knowledge presented indigenous peoples' knowledge	No	No	Select chapters	Select chapters
Knowledges treated as situated and unique	No	No	Select chapters	Select chapters
Value assigned to indigenous knowledges	No	No	Select chapters	Select chapters
Inclusion of indigenous peoples' in research process	No	No	No	Select chapters
Indigenous peoples' generated knowledge	No	No	No	Select chapters

United Nations Framework Convention on Climate Change (UNFCCC)

IPCC (AR4)

Arctic Climate Impact Assessment (ACIA) - <https://acia.amap.no/>

Smith and Sharp (2012)

UNFCCC (COP28 and associated acts ...)

Acknowledging that climate change is a common concern of humankind, Parties should, when taking action to address climate change, respect, promote and consider their respective obligations on human rights, the right to health, the rights of **indigenous** peoples, local communities, migrants, children, persons with disabilities and people in vulnerable situations and the right to development, as well as gender equality, empowerment of women and intergenerational equity,

Paris Agreement (2 mentions of the word 'Indigenous')

Presence of Indigenous Knowledge holders at COP events

<http://www.iipfcc.org/>

Increasingly question of justice (past and present)



Loss and damage / Climate Justice

How to assess the impacts of climate change (tangible and intangible) and who will pay?

Bali Principles on Climate Justice (2002, <https://www.corpwatch.org/article/bali-principles-climate-justice>)

- 3. *Climate Justice affirms the rights of indigenous peoples and affected communities to represent and speak for themselves.*
- 5. *Climate Justice demands that communities, particularly affected communities play a leading role in national and international processes to address climate change.*

Societal 'tipping point' (parallel to climate science; i.e. a point of no return):

3 | KINSHIP AND CLIMATE CHANGE

One lesson I want to draw on here is that the dangers many indigenous peoples face due to climate change are rooted in the relational qualities they have with other societal institutions. These qualities, among others, are the elements of mutual responsibility in coordinated action. In contemporary indigenous intellectual traditions and political activism, relationships of mutual responsibilities, infused with appropriate qualities like consent and reciprocity, are often referred to as kin relationships. *Kincentric* perspectives on climate change can suggest strong reasons for why it may be too late to address dangerous climate change, whether in the sense of stopping a 2°C rise or averting climate injustices.

Whyte (2019)

IPCC Report(s)

Multiple reports that come out continuously (Working Group Reports, Special Reports, Assessment Reports (currently AR6), Synthesis Reports)

Still relatively scattered mentions of 'Indigenous' or 'ILK'

Appreciation of importance with respect to adaptation

Meaningful participation and inclusive planning, informed by cultural values, Indigenous Knowledge, local knowledge, and scientific knowledge can help address adaptation gaps and avoid maladaptation (*high confidence*). Such actions with flexible pathways may encourage low-regret and timely actions (*very high confidence*). Integrating climate adaptation into social protection programmes, including cash transfers and public works programmes, would increase resilience to climate change, especially when supported by basic services and infrastructure (*high confidence*). {WGII SPM C.2.3, WGII SPM C.4.3, WGII SPM C.4.4, WGII SPM C.2.9, WGII WPM D.3}

IPCC Synthesis Reports Climate Change (2023)

IPCC language

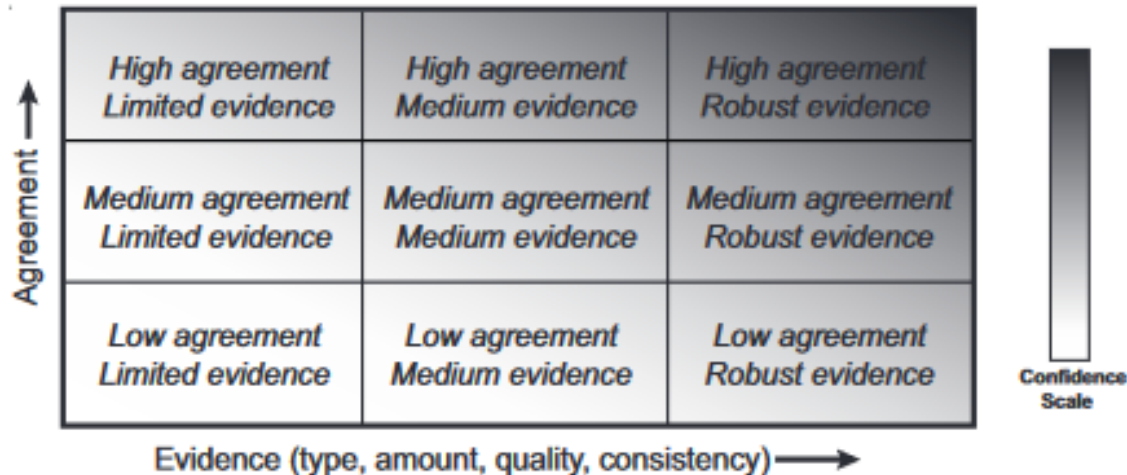


Figure 1: A depiction of evidence and agreement statements and their relationship to confidence. Confidence increases towards the top-right corner as suggested by the increasing strength of shading. Generally, evidence is most robust when there are multiple, consistent independent lines of high-quality evidence.

Table 1. Likelihood Scale	
Term*	Likelihood of the Outcome
Virtually certain	99-100% probability
Very likely	90-100% probability
Likely	66-100% probability
About as likely as not	33 to 66% probability
Unlikely	0-33% probability
Very unlikely	0-10% probability
Exceptionally unlikely	0-1% probability

* Additional terms that were used in limited circumstances in the AR4 (*extremely likely* – 95-100% probability, *more likely than not* – >50-100% probability, and *extremely unlikely* – 0-5% probability) may also be used in the AR5 when appropriate.

IPCC AR6 Mountain Chapter

quantification of **Contribution of climate change** (how much is climate change to blame)

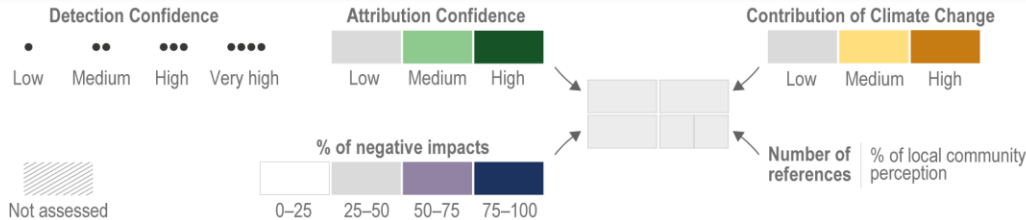
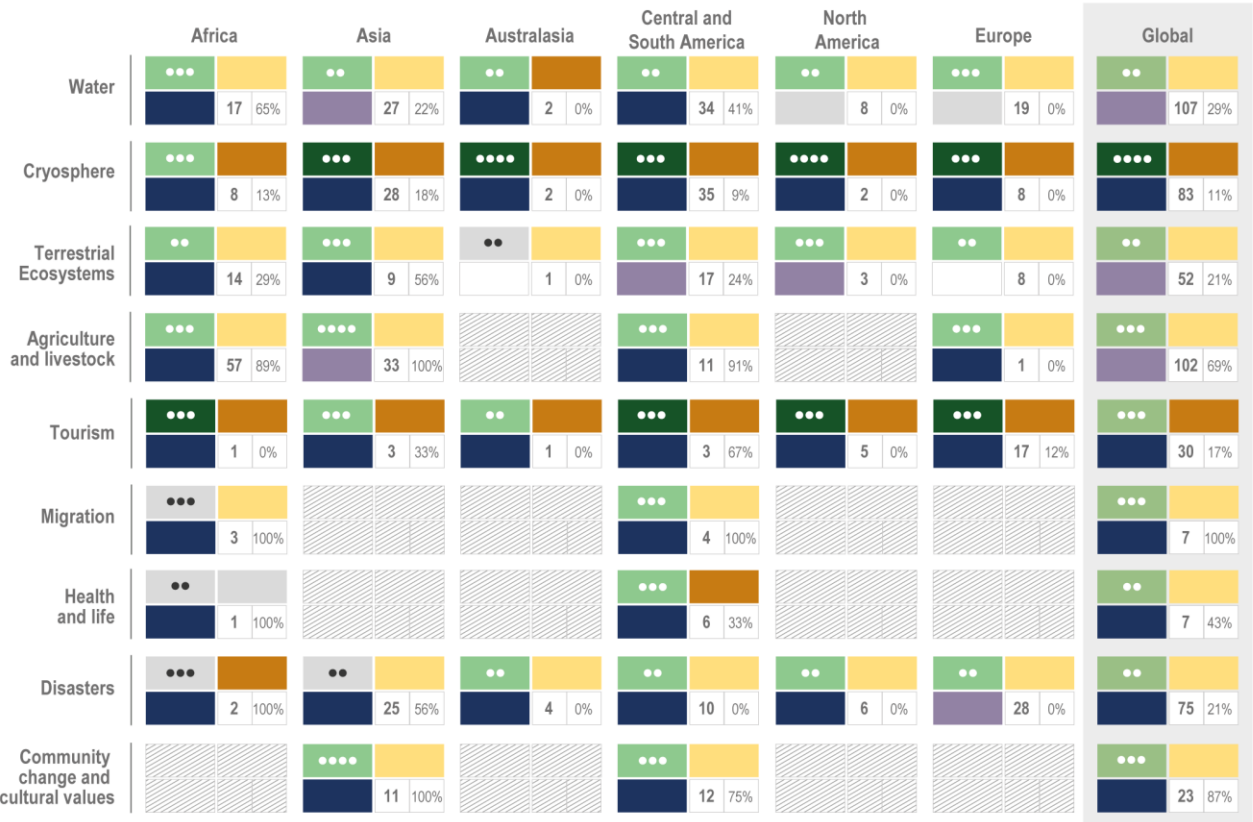
quantification of **Attribution confidence** (can impacts be directly linked to anthropogenic climate change)

Detection confidence (can we even measure it?)

Type of impact (negative/positive)

Percentage of local community input

Detection and attribution of observed impacts of anthropogenic climate change in mountain regions



IPCC AR6 Mountain Chapter

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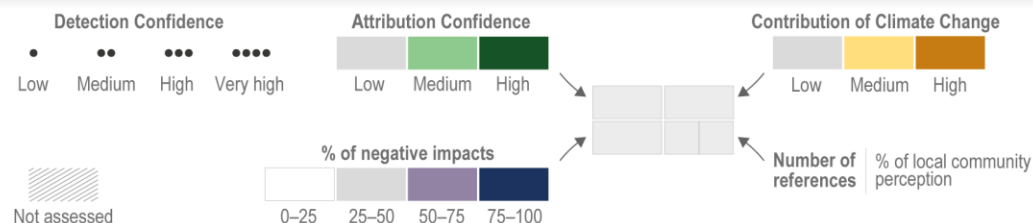
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Detection and attribution of observed impacts of anthropogenic climate change in mountain regions



6 December / Day 2

Welcome to our second session 😊

Learning Objectives: Understand what role ILK has played in the Arctic and mountain regions specifically. Appreciate how ILK can be combined with 'conventional' science when addressing climate change, but also recapitulate on the gaps we are still facing.

Roadmap for today:

- 9:00-9:30 Catch up from Session 1
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- 10:15-10:30 **Break**
- 10:30-11:30 [Watch-Think-Answer](#) (in-class group discussion)
- 11:30-12:00 ILK in Mountain contexts
- 12:00-13:00 **Lunch Break**
- 13:00-13:45 ILK and Disaster Risk Reduction (Read-Think-Discuss, in-class group discussion)
- 13:45-14:00 **Break**
- 14:00-14:45 International efforts to integrate ILK
- 14:45-15:00 **Break**
- 15:00-16:00 ILK in the Arctic - Power Dynamics - Guest Speaker (Anna-Maria Walter)
- 16:00-16:15 **Break**
- 16:15-17:00 ILK in Global Assessments

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AMAP (Arctic Monitoring and Assessment Programme)

ACIA Report (2004) - <https://acia.amap.no/>

- refers to Indigenous knowledge and impacts in all chapters
- has a dedicated chapter and key finding specifically referring to ILK



8. Indigenous communities are facing major economic and cultural impacts.

- Many Indigenous Peoples depend on hunting polar bear, walrus, seals, and caribou, herding reindeer, fishing, and gathering, not only for food and to support the local economy, but also as the basis for cultural and social identity.
- Changes in species' ranges and availability, access to these species, a perceived reduction in weather

predictability, and travel safety in changing ice and weather conditions present serious challenges to human health and food security, and possibly even the survival of some cultures.

- Indigenous knowledge and observations provide an important source of information about climate change. This knowledge, consistent with complementary information from scientific research, indicates that substantial changes have already occurred.

AMAP (Arctic Monitoring and Assessment Programme)

"There used to be different layers of snow back then. The wind would not blow as hard, not make the snow as hard as it is now... It's really hard to make shelters with that kind of snow because it's usually way too hard right down to the ground."

- T. Qaqimat, Baker Lake, Canada, 2001

"Change has been so dramatic that during the coldest month of the year, the month of December 2001, torrential rains have fallen in the Thule region so much that there appeared a thick layer of solid ice on top of the sea ice and the surface of the land... which was very bad for the paws of our sled dogs."

- Uusaqqak Qujaukitsoq, Qaanaaq, Greenland, 2002

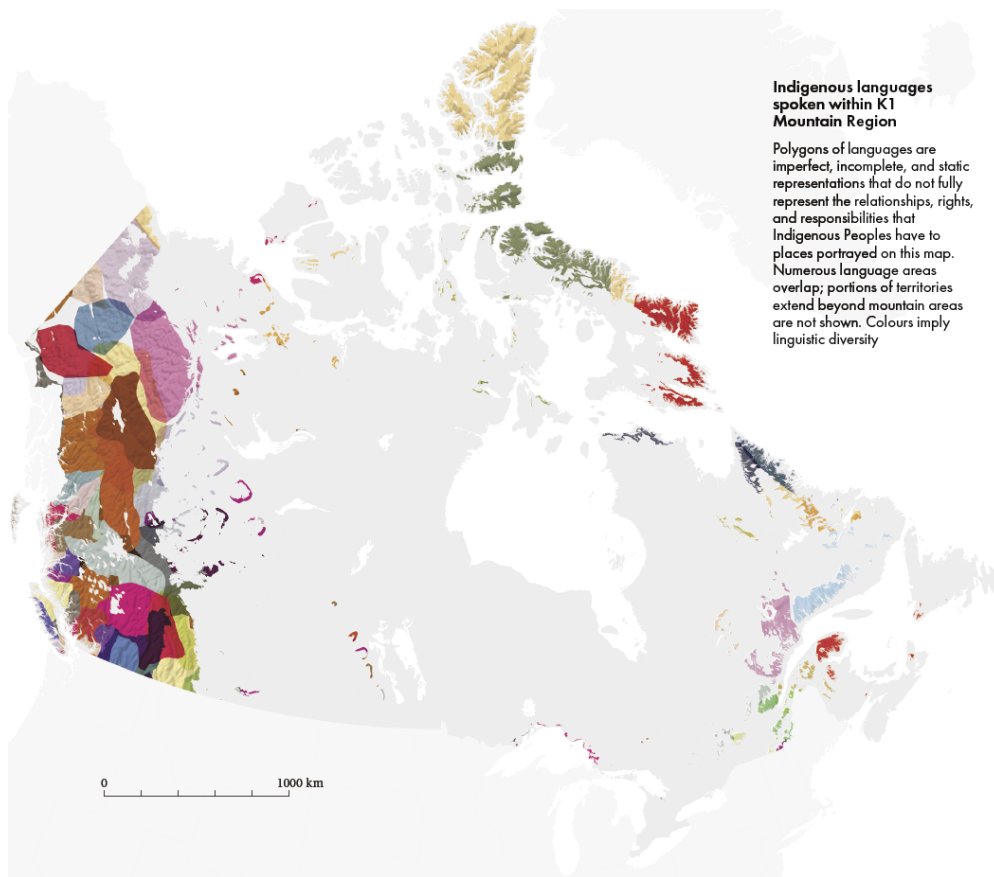
"It used to be that there would be proper freezing which would dry the lichen and the snow would fall on top. There would be rain that would form the bottom, which would then freeze properly. Now it rains, and the bottom freezes wet, and this is bad for the reindeers. It ruins the lichen. Ice is everywhere and the reindeer cannot get through. This has meant death to a number of reindeers because they cannot get to the lichen."

- Niila Nikodemus, 86, the oldest reindeer herder in Purnumukka, Finland, 2002

"First it snows, then it melts, like it would be summertime. And this all over again. First there is a big snowfall, then it warms up and then it freezes. During winter now it can rain, as happened last New Year. Before it never rained during wintertime. Rain in the middle of winter? To the extent that snow disappears? Yes, it is true. Rain, and snow melts!"

- Vladimir Lifov, Lovozero, Russia, 2002

Canadian Mountain Assessment



The Canadian Mountain Assessment

WALKING TOGETHER TO ENHANCE UNDERSTANDING
OF MOUNTAINS IN CANADA

Graham McDowell, Madison Stevens, Shawn Marshall, et al.



<https://ucp.manifoldapp.org/projects/9781773855103>

Canadian Mountain Assessment

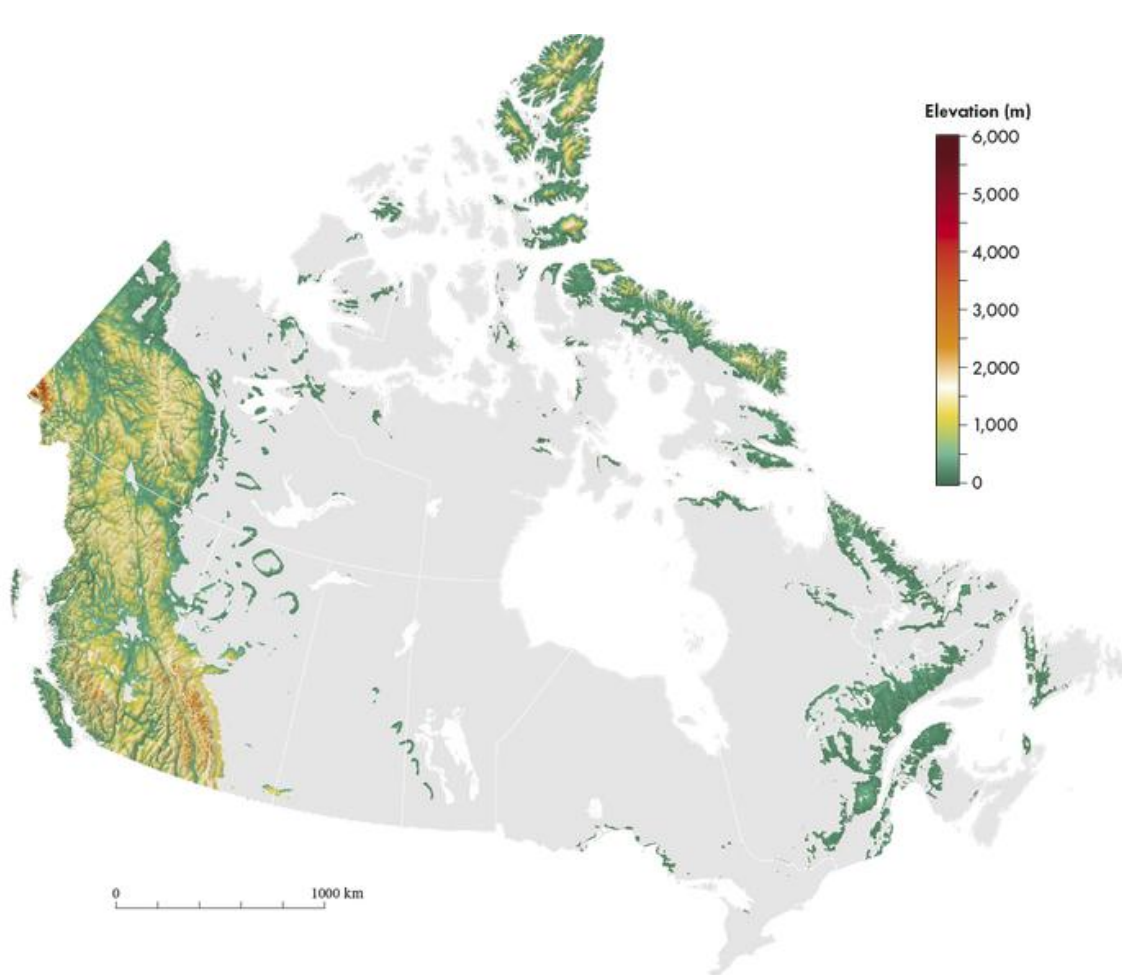


Figure 1.1: Mountainous areas in Canada based on McDowell & Guo (2021), following definition given by Kapos et al. (2000).

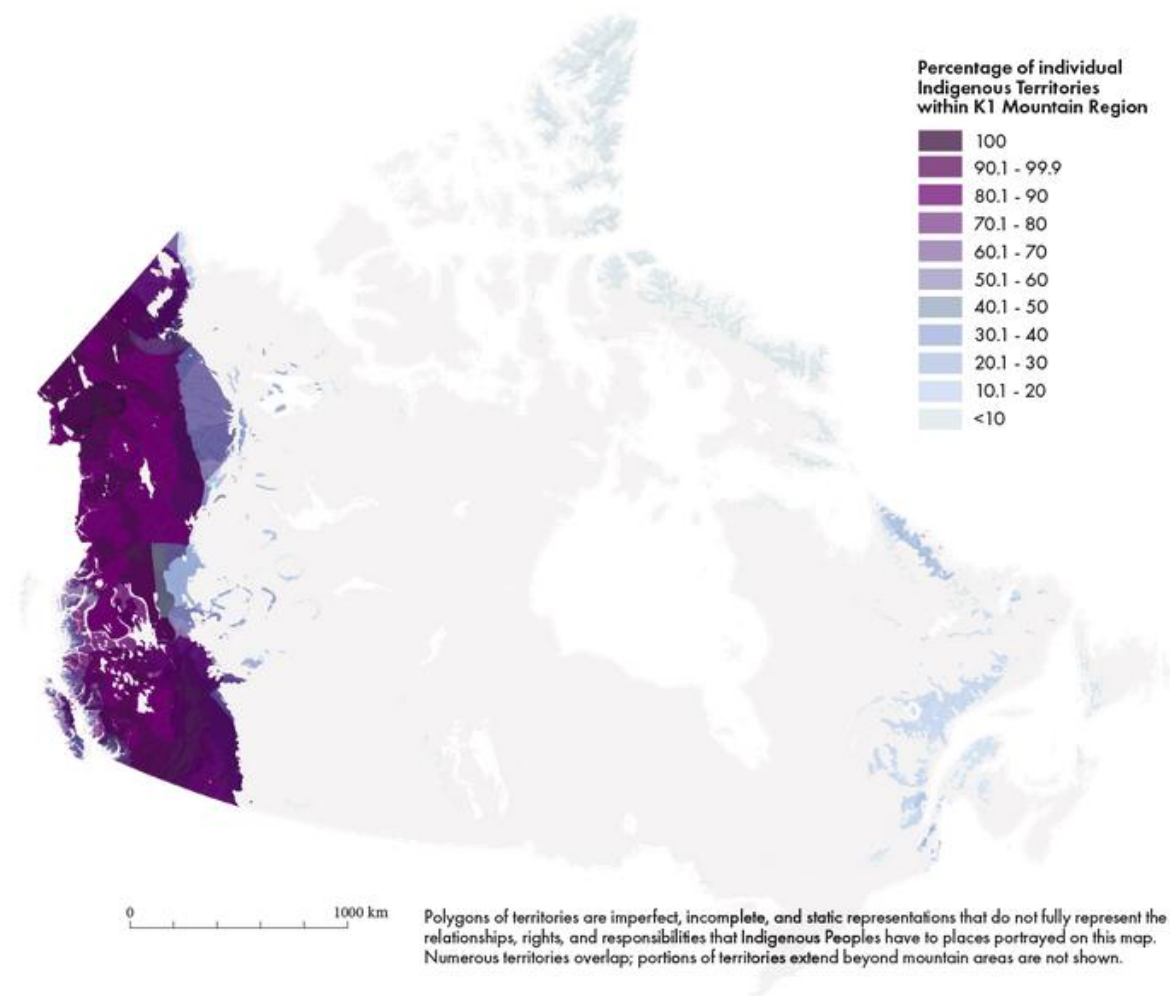
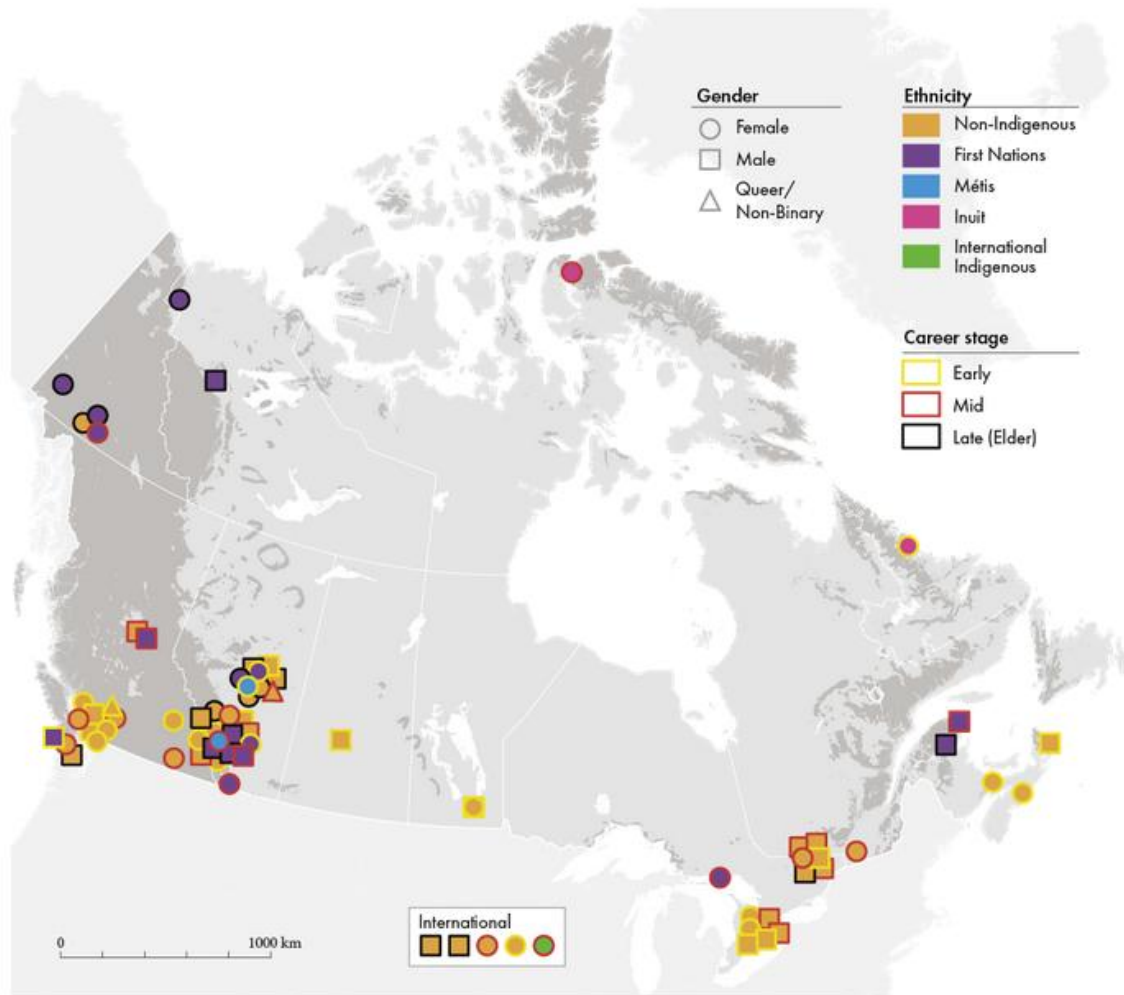


Figure 1.2: Indigenous territories associated with mountain areas in Canada. Based on McDowell & Guo (2021). Data from Native-Land.ca.

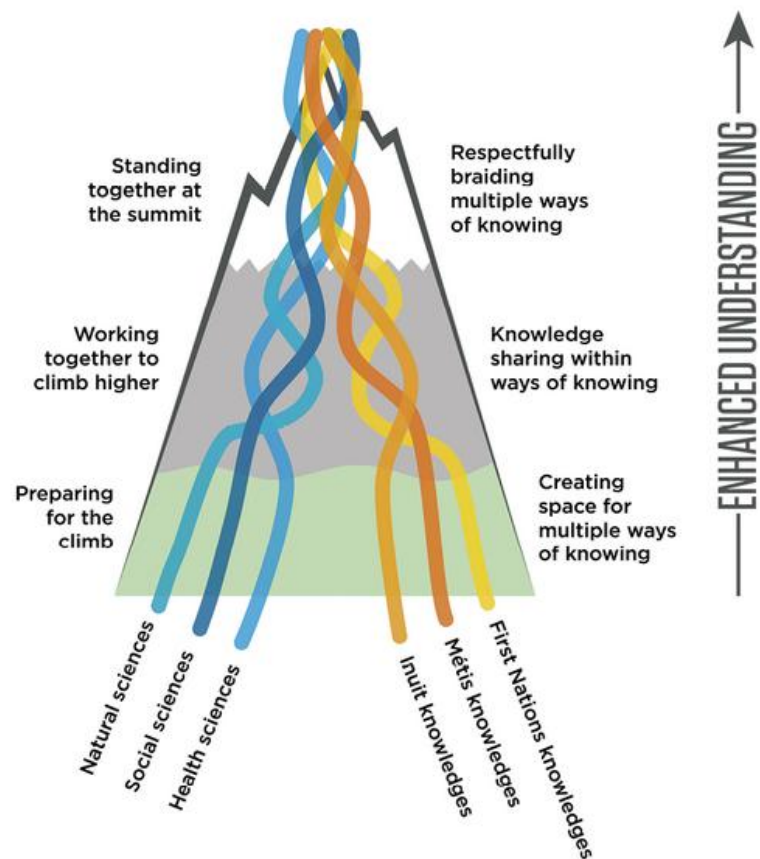
Canadian Mountain Assessment



Not just an assessment of Indigenous and local knowledge but a concrete inclusion in the process of assessment itself.

Figure 1.8: CMA contributors by First Nations, Métis, and Inuit/non-Indigenous identity; career stage; location; and gender (does not include external reviewers).

Canadian Mountain Assessment



Not just an assessment of Indigenous and local knowledge but a concrete inclusion in the process of assessment itself.

More inclusive 'assessment' process

Figure 1.9: Simplified depiction of a multiple evidence base approach in the context of the CMA. Much like climbing to the summit of a mountain, respectfully braiding ways of knowing is an aspiration, but such an outcome is not guaranteed. The success of the effort can be determined not only by the point the climbers reach, but also by the learning along the way.

Canadian Mountain Assessment



*Pnnal Bernard Jerome, Micmacs of
Gesgapegiag, 2022, LC 1.1*



*Hayden Melting Tallow, Siksika
Nation, Blackfoot Confederacy,
2022, LC 1.2*



Not just an assessment of Indigenous and local knowledge but a concrete inclusion in the process of assessment itself.

More inclusive 'assessment' process

Innovative way of presenting and communication

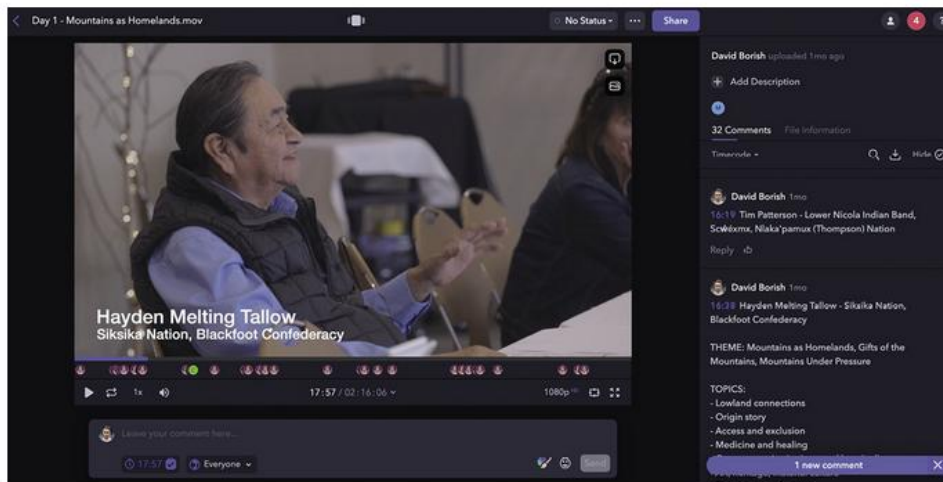


Figure 1.12: Learning Circle video with searchability according to participant name, community and/or Nation, themes, and topics

Alternative framing of knowledges (CMA)

Chapter 3. Mountains as Homelands

The Mountains as Homelands chapter considers how mountains in Canada are experienced and shaped as Homelands by Indigenous Peoples and homes by non-Indigenous people. The chapter

Chapter 4. Gifts of the Mountains

The Gifts of the Mountains chapter explores the contributions of mountains to the wellbeing of human communities. It uses the framing of gifts as an alternative to the conventional descriptions of resources or ecosystem services, and reveals how, for many people in Canada, mountains provide material, artistic, pedagogical, emotional, and spiritual gifts. The chapter also discusses how particular users and communities receive benefits derived from energy, minerals, and forests found in mountains. Importantly, the chapter calls attention to the idea that many gifts from mountains are situated in reciprocal relationships where users receive foods, medicines, water, or recreational space, as personal gifts which, in turn, inspire wonder, awe, respect, and care. Such reciprocity is often, but not exclusively, associated with Indigenous worldviews. Ultimately, the

Thank you

